

KNOWLEDGE INSTITUTE OF TECHNOLOGY

(An Autonomous Institution)

Approved by AICTE, Affiliated to Anna University, Chennai.

Accredited by NBA (CSE, ECE, EEE & MECH), Accredited by NAAC with "A" Grade
KIOT Campus, Kakapalayam (PO), Salem – 637 504, Tamil Nadu, India.



B.E. / B.Tech. Regulations 2023

B.E. – Electronics and Computer Engineering

Semester – II

Curriculum and Syllabi

(For the Students admitted from the Academic Year 2025-26 Onwards)

For the Students admitted from the Academic Year 2025-26 Onwards
Knowledge Institute of Technology
KIOT Campus, Kakapalayam,
Salem - 637 504

Version: 1.0

Date: 15.12.2025

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6/1/25

U. Lip

CHAIRPERSON

Board of Studies
Faculty of Electronics & Communication Engg
Knowledge Institute of Technology
KIOT Campus, Kakapalayam,
Salem - 637 504

	KNOWLEDGE INSTITUTE OF TECHNOLOGY(AUTONOMOUS), SALEM -637504
	Approved by AICTE, Affiliated to Anna University, Accredited by NAAC and NBA (B.E.:Mech., ECE, EEE & CSE)
	Website: www.kiot.ac.in

B.E. / B.Tech. REGULATIONS 2023 (R 2023)
CHOICE BASED CREDIT SYSTEM AND OUTCOME BASED EDUCATION

B.E. ELECTRONICS AND COMPUTER ENGINEERING

VISION OF THE INSTITUTE

To be a world class institution to impart value and need based professional education to the aspiring youth and carving them into disciplined world class professional who have the quest for excellence, achievement orientation and social responsibilities

MISSION OF THE INSTITUTE

A	To promote academic growth by offering state-of-art undergraduate, postgraduate, and doctoral programs and to generate new knowledge by engaging in cutting – edge research
B	To nurture talent, innovation, entrepreneurship, all-round personality, and value system among the students and to foster competitiveness among students
C	To undertake collaborative projects which offer opportunities for long-term interaction with academia and industry
D	To pursue global standards of excellence in all our endeavors namely teaching, research, consultancy, continuing education and support functions

VISION OF THE DEPARTMENT

To produce competent, industry-relevant engineering professionals, blending expertise in electronic hardware and software development with strong ethical values and social responsibility

MISSION OF THE DEPARTMENT

M1	To provide quality technical education in Electronics and Computer Engineering through well-equipped infrastructure and innovative teaching
M2	To equip students with industry relevant skills and emerging technologies by collaborating with industries
M3	To foster professional culture, research aptitude and entrepreneurial abilities among students

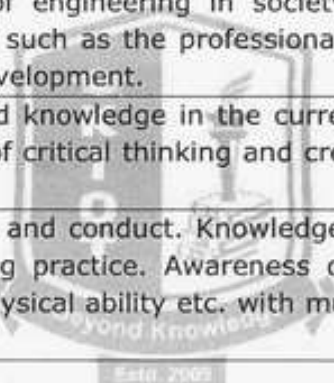
PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

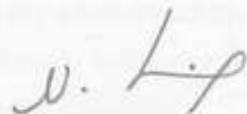
PEO 1	Graduates will demonstrate strong technical skills in digital computing technologies and software systems
PEO 2	Graduates will engage in professional practice and apply expertise in cyber physical systems to solve real world engineering problems
PEO 3	Graduates will demonstrate professional culture, research aptitude and entrepreneurial skills to drive technological innovation

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WASHINGTON ACCORD KNOWLEDGE AND ATTITUDE PROFILE (WK)	
WK 1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK 2	Conceptually based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK 3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK 4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK 5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK 6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
WK 7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
WK 8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
WK 9	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.




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PROGRAM OUTCOMES (POs)	
Engineering Graduates will be able to:	
PO1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO2	Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4).
PO3	Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5).
PO4	Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
PO5	Modern Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6).
PO6	The Engineer and Society: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
PO7	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9).
PO8	Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9	Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
PO10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11	Life-Long Learning: Recognize the need for and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8).

PROGRAM SPECIFIC OUTCOMES (PSOs)	
After the successful completion of B.E. Programme in Electronics and Computer Engineering, the graduates will be able to	
PSO 1	Apply concepts of digital architecture to build efficient hardware- software integrated solutions.
PSO 2	Develop and analyze software applications using embedded platforms and operating system frameworks for reliable computing solutions.
PSO 3	Develop and analyze cyber physical systems to address contemporary engineering challenges.



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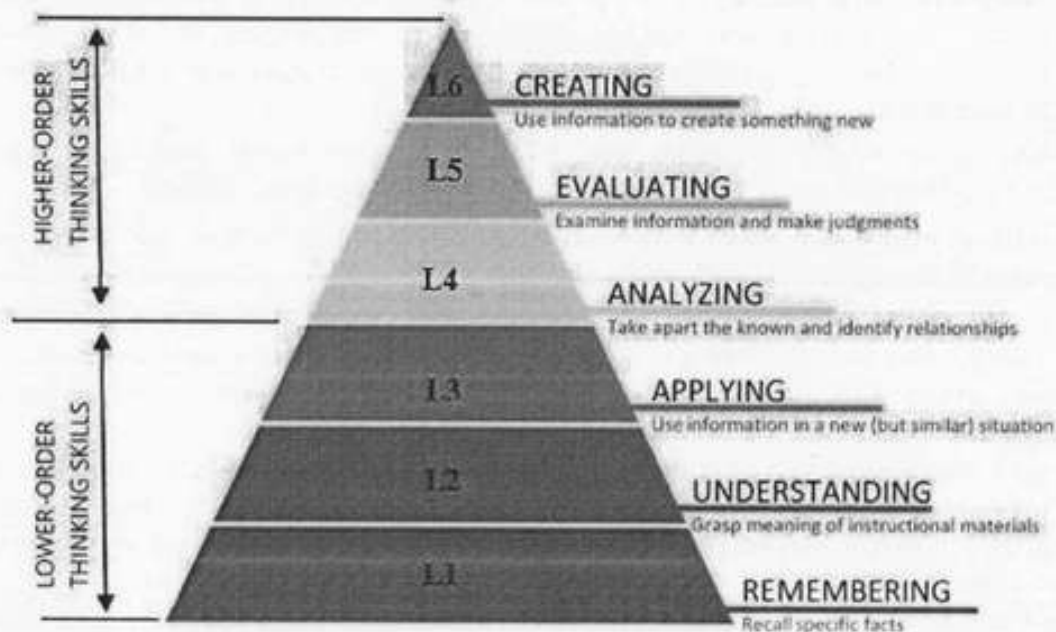
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BLOOM'S TAXONOMY LEVELS AND ACTION VERBS

(A) BLOOM'S TAXONOMY LEVELS (BTL):

Bloom's Taxonomy (BT) is based on the belief that learners must first acquire basic foundational knowledge about a subject before progressing to more complex types of thinking, such as analysis and evaluation. Bloom's Taxonomy levels help faculty to guide students through the learning process, from fundamental remembering and understanding to more complex evaluating and creating.



At KIOT, Curriculum Design, Delivery and Assessment (CDDA) are carried out based on the Blooms' Taxonomy Levels (BTL). Its organized set of objectives helps teachers to plan and deliver appropriate instruction, design valid assessment tasks and schemes, and ensure that instruction and assessment are aligned with the objectives.

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(B) BLOOM'S TAXONOMY ACTION VERBS

I. Remembering	II. Understanding	III. Applying	IV. Analyzing	V. Evaluating	VI. Creating
Exhibit memory of previously learned material by recalling facts, terms, basic concepts, and answers.	Demonstrate understanding of facts and ideas by organizing, comparing, interpreting, giving descriptions, and stating main ideas.	Solve problems to new situations by applying acquired knowledge, facts, techniques and rules in a different way.	Examine and break information into parts by identifying motives or causes. Make inferences and find evidence to support generalizations.	Present and defend opinions by making judgments about information, validity of ideas, or quality of work based on a set of criteria.	Compile information together in a different way by combining elements in a new pattern or proposing new Solutions.
- Choose - Define - Find - How - Label - List - Match - Name - Omit - Recall - Relate - Select - Show - Spell - Tell - What - When - Where - Which - Who - Why	- Classify - Compare - Contrast - Demonstrate - Explain - Extend - Illustrate - Infer - Interpret - Outline - Relate - Rephrase - Show - Summarize - Translate	- Apply - Build - Choose - Construct - Develop - Experiment with - Identify - Interview - Make use of - Model - Organize - Plan - Select - Solve - Utilize	- Analyze - Assume - Categorize - Classify - Compare - Conclusion - Contrast - Discover - Dissect - Distinguish - Divide - Examine - Function - Inference - Inspect - List - Motive - Relationships - Simplify - Survey - Take part in - Test for - Theme	- Agree - Appraise - Assess - Award - Choose - Compare - Conclude - Criteria - Criticize - Decide - Deduct - Defend - Determine - Disprove - Estimate - Evaluate - Explain - Importance - Influence - Interpret - Judge - Justify - Mark - Measure - Opinion - Perceive - Prioritize - Prove - Rate - Recommend - Rule on - Select - Support - Value	- Adapt - Build - Change - Choose - Combine - Compile - Compose - Construct - Create - Delete - Design - Develop - Discuss - Elaborate - Estimate - Formulate - Happen - Imagine - Improve - Invent - Make up - Maximize - Minimize - Modify - Original - Originate - Plan - Predict - Propose - Solution - Solve - Suppose - Test - Theory

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KNOWLEDGE INSTITUTE OF TECHNOLOGY (AUTONOMOUS), SALEM - 637504											
B.E. ELECTRONICS AND COMPUTER ENGINEERING									Version: 1.0		
Courses of Study and Scheme of Assessment (Regulations 2023)									Date: 15.12.2025		
Sl. No.	Course Code	Course Title	Periods / Week						Maximum Marks		
			CAT	CP	L	T	P	C	IA	ESE	Total
SEMESTER I											
	BE23IP001	Induction Programme	IP	80	60	-	20	NC	-	-	-
THEORY											
1	BE23EN101	Communicative English-I	HS	2	1	1	0	2	40	60	100
2	BE23MA201	Calculus for Engineers	BS	3	2	1	0	3	40	60	100
3	BE23PH204	Engineering Physics	BS	3	3	0	0	3	40	60	100
4	BE23CY201	Engineering Chemistry	BS	3	3	0	0	3	40	60	100
5	BE23GE301	Overview of Engineering and Technology	ES	3	3	0	0	3	40	60	100
6	BE23MC901	தமிழர் மரபு / Heritage of Tamils	MC	1	1	0	0	1	40	60	100
THEORY CUM PRACTICAL											
7	BE23GE306	Problem Solving and C Programming	ES	5	3	0	2	4	50	50	100
PRACTICAL											
8	BE23BS201	Physics and Chemistry Laboratory	BS	4	0	0	4	2	60	40	100
9	BE23GE305	Engineering Practices Laboratory	ES	4	0	0	4	2	60	40	100
EMPLOYABILITY ENHANCEMENT											
10	BE23PT801	Human Excellence and Value Education - I	EEC	2	1	0	1	NC	100	-	100
Total				30	17	2	11	23	510	490	1000
SEMESTER II											
THEORY											
1	BE23EN102	Communicative English-II	HS	2	1	1	0	2	40	60	100
2	BE23MA210	Linear Algebra and Numerical Methods	BS	3	2	1	0	3	40	60	100
3	BE23GE303	Engineering Graphics and Circuit Drawings	ES	5	1	0	4	3	40	60	100
4	BE23EC401	Electronic Devices	PC	3	3	0	0	3	40	60	100
5	BE23MC902	தமிழரும் தொழில்நுட்பமும் / Tamils and Technology	MC	1	1	0	0	1	40	60	100
6	BE23MC903	Universal Human Values and Ethics	MC	3	2	1	0	3	40	60	100
THEORY CUM PRACTICAL											
7	BE23GE310	Object Oriented Programming using C++	ES	5	3	0	2	4	50	50	100
8	BE23EC402	Circuit Theory and Analysis	PC	5	3	0	2	4	50	50	100
EMPLOYABILITY ENHANCEMENT											
9	BE23PT802	Human Excellence and Value Education-II	EEC	2	1	0	1	NC	100	-	100
10	BE23PT804	Engineering Clinic-I	EEC	2	0	0	2	1	100	-	100
11	BE23PT806	Aptitude Skills-I	EEC	1	0	0	1	0.5	100	-	100
Total				32	17	3	12	24.5	640	460	1100

w.e.f. 22/12/2025

Passed in BoS Meeting held on 06/12/2025

Approved in Academic Council Meeting held on 15/12/2025

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KNOWLEDGE INSTITUTE OF TECHNOLOGY (AUTONOMOUS), SALEM - 637504

B.E. ELECTRONICS AND COMPUTER ENGINEERING

Courses of Study and Scheme of Assessment (Regulations 2023)

Sl. No.	Course Code	Course Title	Periods / Week						Maximum Marks		
			CAT	CP	L	T	P	C	IA	ESE	Total
SEMESTER III											
	THEORY										
1	BE23MA203	Discrete Mathematics	BS	3	2	1	0	3	40	60	100
2	BE23EX401	Computer Organization and Operating Systems	PC	3	2	1	0	3	40	60	100
	THEORY CUM PRACTICAL										
3	BE23CS310	Fundamentals of Data Structures and Database	ES	5	2	1	2	4	50	50	100
4	BE23EX402	Analog Integrated Circuits	PC	5	2	1	2	4	50	50	100
5	BE23EX403	Digital Electronics and Microprocessor	PC	5	2	1	2	4	50	50	100
6	BE23EX404	Python Programming	PC	5	2	1	2	4	50	50	100
	PRACTICAL										
7	BE23EN103	Professional Communication Laboratory-I	HS	2	0	0	2	1	60	40	100
	EMPLOYABILITY ENHANCEMENT										
8	BE23PT805	Engineering Clinic-II	EEC	2	0	0	2	1	100	-	100
9	BE23PT807	Aptitude Skills-II	EEC	1	0	0	1	0.5	100	-	100
		Total		31	12	6	13	24.5	540	360	900
SEMESTER IV											
	THEORY										
1	BE23MA206	Mathematics for Business Analytics	BS	3	2	1	0	3	40	60	100
2	BE23EX405	Digital Communication	PC	3	2	1	0	3	40	60	100
3	BE23EX406	Signal Processing for Discrete Systems	PC	3	2	1	0	3	40	60	100
4	BE23CS407	Design and Analysis of Algorithms	PC	3	2	1	0	3	40	60	100
5	BE23MC904	Environmental Science and Sustainability	MC	2	2	0	0	NC	100	-	100
6	BE23MC906	Entrepreneurship and Start-ups	PW	3	2	1	0	NC	100	-	100
	THEORY CUM PRACTICAL										
7	BE23CS315	JAVA Programming	ES	5	2	1	2	4	50	50	100
8	BE23EX407	Microcontroller and Embedded Systems	PC	5	2	1	2	4	50	50	100
	PRACTICAL										
9	BE23EN104	Professional Communication Laboratory-II	HS	2	0	0	2	1	60	40	100
10	BE23GE311	Design Thinking	ES	4	0	0	4	2	100	-	100
	EMPLOYABILITY ENHANCEMENT										
11	BE23PT808	Aptitude Skills-III	EEC	1	0	0	1	0.5	100	-	100
		Total		34	16	7	11	23.5	720	380	1100

w.e.f. 22/12/2025

Passed in BoS Meeting held on 06/12/2025

Approved in Academic Council Meeting held on 15/12/2025

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KNOWLEDGE INSTITUTE OF TECHNOLOGY (AUTONOMOUS), SALEM - 637504												
B.E. ELECTRONICS AND COMPUTER ENGINEERING												
Courses of Study and Scheme of Assessment (Regulations 2023)												
Sl. No.	Course Code	Course Title	Periods / Week						Maximum Marks			
			CAT	CP	L	T	P	C	IA	ESE	Total	
SEMESTER V												
	THEORY											
1	BE23EX408	Computer Networks and Security	PC	3	2	1	0	3	40	60	100	
2	BE23EX409	Cyber-Physical Systems	PC	3	2	1	0	3	40	60	100	
3	BE23XX6XX	Open Elective - I**	OE	3	2	1	0	3	40	60	100	
4	BE23AC905	Indian Constitution	AC	2	2	0	0	NC	100	-	100	
	THEORY CUM PRACTICAL											
5	BE23EX5XX	Professional Elective - I	PE	5	2	1	2	4	50	50	100	
6	BE23EX5XX	Professional Elective - II	PE	5	2	1	2	4	50	50	100	
	PRACTICAL											
7	BE23PW701	Make-A-Product- PhaseI*	PW	1	0	0	1	NC	-	-	-	
	EMPLOYABILITY ENHANCEMENT											
8	BE23PT809	Aptitude Skills-IV	EEC	1	0	0	1	0.5	100	-	100	
9	BE23PT810	Coding Skills-I	EEC	2	0	0	2	1	100	-	100	
10	BE23PT812	Technical Comprehension and Mock Interview-I	EEC	1	0	0	1	0.5	100	-	100	
Total				26	12	5	9	19	620	280	900	
SEMESTER VI												
	THEORY											
1	BE23EX410	Design Principles of Application Software	PC	3	2	1	0	3	40	60	100	
2	BE23XX6XX	Open Elective - II**	OE	3	2	1	0	3	40	60	100	
3	BE23MC907	Disaster Management and Preparedness	MC	2	2	0	0	NC	100	-	100	
	THEORY CUM PRACTICAL											
4	BE23GE312	AI and IoT for Engineering Applications	ES	5	3	0	2	4	50	50	100	
5	BE23EX5XX	Professional Elective - III	PE	5	2	1	2	4	50	50	100	
6	BE23EX5XX	Professional Elective - IV	PE	5	2	1	2	4	50	50	100	
	PRACTICAL											
7	BE23PW702	Make-A-Product- PhaseII	PW	2	0	0	2	1	100	-	100	
	EMPLOYABILITY ENHANCEMENT											
8	BE23PT803	Human Excellence and Value Education-III	EEC	2	1	0	1	NC	100	-	100	
9	BE23PT811	Coding Skills-II	EEC	2	0	0	2	1	100	-	100	
10	BE23PT813	Technical Comprehension and Mock Interview-II	EEC	1	0	0	1	0.5	100	-	100	
Total				30	14	4	12	20.5	730	270	1000	

w.e.f. 22/12/2025

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B.E. ELECTRONICS AND COMPUTER ENGINEERING											
Courses of Study and Scheme of Assessment (Regulations 2023)											
Sl. No.	Course Code	Course Title	Periods / Week						Maximum Marks		
			CAT	CP	L	T	P	C	IA	ESE	Total
SEMESTER VII											
	THEORY										
1	BE23HS105	Project Management and Finance	HS	3	2	1	0	3	40	60	100
2	BE23EX411	Hardware-Software Co-Design	PC	3	2	1	0	3	40	60	100
3	BE23XX6XX	Open Elective - III**	OE	3	2	1	0	3	40	60	100
	THEORY CUM PRACTICAL										
4	BE23EX5XX	Professional Elective - V	PE	5	2	1	2	4	50	50	100
5	BE23EX5XX	Professional Elective - VI	PE	5	2	1	2	4	50	50	100
	PRACTICAL										
6	BE23EX702	Project Work - Phase I	PW	3	0	0	3	1	100	-	100
	EMPLOYABILITY ENHANCEMENT										
7	BE23PT814	Industrial Training / Entrepreneurship / Undergraduate Research Activity / Company Certification	EEC	6	0	0	6	3	100	-	100
Total				28	10	5	13	21	420	280	700
SEMESTER VIII											
	PRACTICAL										
1	BE23EX703	Project Work - Phase II	PW	20	0	0	20	9	60	40	100
Total				20	0	0	20	9	60	40	100
									Total Number of Credits: 165		
* The course will be considered for grading in semester VI only.											
** Student shall choose a course from the list of open electives offered by other departments or from emerging technology verticals including online courses with prior approval.											

w.e.f. 22/12/2025
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SEMESTER-WISE CREDITS DISTRIBUTION

SUMMARY											
Sl. No.	Course Category	Credits per Semester								Credits	Credit %
		I	II	III	IV	V	VI	VII	VIII		
1	HS	2	2	1	1	-	-	3	-	9	5
2	BS	11	3	3	3	-	-	-	-	20	12
3	ES	9	7	4	6	-	-	-	-	26	16
4	PC	-	7	15	13	6	7	3	-	51	31
5	PE	-	-	-	-	8	8	8	-	24	15
6	OE	-	-	-	-	3	3	3	-	9	5
7	PW	-	-	-	-	✓	1	1	9	11	7
8	EEC	✓	1.5	1.5	0.5	2	1.5	3	-	10	6
9	MC/NC/AC	1	4	-	✓	✓	-	-	-	5	3
	Total	23	24.5	24.5	23.5	19	20.5	21	9	165	100

CAT	Category of Course	IP	Induction Programme	PW	Project Work Courses
CP	Contact Periods	HS	Humanities, Social Sciences and Management Courses	EEC	Employability Enhancement Courses
L	Lecture Hours	BS	Basic Science Courses	MC/NC/AC	Mandatory Courses/Non-Credit Courses/Audit Courses
T	Tutorial Hours	ES	Engineering Science Courses	IA	Internal Assessment
P	Practical Hours	PC	Professional Core Courses	ESE	Semester End Examination
C	Credits	PE	Professional Elective Courses	AC	Audit Courses
				OE	Open Elective Courses



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BE23EN101	COMMUNICATIVE ENGLISH - I	CP	L	T	P	C
		2	1	1	0	2
Programme & Branch	COMMON TO ALL B.E./B.TECH. BRANCHES	Version: 1.1				
Course Objectives:						
1.	To enable learners to use words appropriately in their communication.					
2.	To enhance learners' grammatical accuracy in communication.					
3.	To develop learners' ability to read and listen to texts in English.					
4.	To strengthen the communication skills of the learners.					
5.	To help learners write appropriately in professional contexts.					
	INTRODUCTION (Not for Examination)					2
Importance	- The course enhances competence in the four modes of literacy: LSRW - It strengthens the learners' presentation skills in various forums. - The course helps the learners to increase their cognitive ability. - It helps learners acquire language competence for campus recruitment.					
Real-life Example(s)	Writing Reports - letter writing - public speaking - presentation - formal and informal conversation					
Linkages	Fundamental knowledge of LSRW Skills in English					
UNIT-I	BASICS OF LANGUAGE					3+3
	Concept: Introduction to Language and Communication - Parts of Speech - Vocabulary: Synonyms & Antonyms, Word formation, Prefixes and Suffixes - One-word substitute - Gerund and Infinitive - Tenses: Simple Present, Present Continuous, Present Perfect, Present Perfect Continuous. Activity: Exercises using worksheets - Word / grammar games - Conducting quiz.					
UNIT-II	LANGUAGE DEVELOPMENT					3+3
	Concept: Tenses: Simple Past, Past Continuous, Simple Future, Future Continuous - Active to Passive Voice - Framing Questions: WH / Yes or No - Modal Verbs - Cause and Effect Expressions - Day to day Idioms & Phrases. Activity: Practice using worksheets - Role play - Face to face conversation.					
UNIT-III	DEVELOPING LISTENING & READING SKILLS					3+3
	Concept: Types of listening - Global accent - Pronunciation, listening to short talks of celebrities, TV shows, announcements, TED Talks - Reading: Skimming and Scanning - Reading Brochures - Understanding sentence structure - Punctuation - News Articles. Activity: Paraphrasing news article - Listening comprehension - Reading comprehension					

w.e.f. 13/10/2025

Passed in BoS Meeting held on 18/09/2025

Approved in Academic Council Meeting held on 11/10/2025

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UNIT – IV	SPEAKING FOR EXPRESSION	3+3
	Concept: Overcoming Mother Tongue Influence - Self-Introduction & Introducing others - Speaking about hobbies, areas of interest, likes and dislikes, Usage of Numerical Adjectives - Relative pronouns - combining sentences using relative pronouns - Discussion on social issues - sharing experience of past and future plans - Talking about engineering devices. Activity: Just a minute talk (JAM) – Debate.	
UNIT-V	TECHNICAL WRITING	3+3
	Concept: Extended definition of Technical Words - Writing abstracts - Note making - Report writing - Techniques of writing a report - Kinds of report - Industrial report - Writing Instructions and recommendations - Formal letters: letter to industry, letter to editor, letter of complaint. Activity: Writing Industrial report - Project report - Technical report.	
	OPEN ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Total (LT)	32 Periods
	Course Outcomes: Upon completion of this course the students will be able to:	BLOOM'S Taxonomy
CO1	Use appropriate words in all kinds of correspondence.	L3 - Apply
CO2	Demonstrate appropriate language use in extended discussions.	L3 - Apply
CO3	Apply the strategies of listening, reading and comprehending the text appropriately.	L3 - Apply
CO4	Construct ideas to be active participants in all kinds of discussions.	L3 - Apply
CO5	Apply technical information and knowledge in practical documents.	L3 - Apply
	TEXTBOOKS:	
1.	Tiwari Anjana, Communication Skills in English, Khanna Publication: New Delhi, 2022.	
	REFERENCE BOOKS:	
1.	Raymond, Murphy, English Grammar in Use, 5 th Edition, Cambridge Press: New York, 2019.	
2.	Wren and Martin, High School English Grammar and Composition, S Chand Publishing: India, 2021.	
3.	Kumar Suresh E, Engineering English, Orient Blackswan, Hyderabad, 2015.	
4.	Kumar Kulbhusan and RS Salaria, Effective Communication Skill, Khanna Publishing House, New Delhi, 2016.	

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	ONLINE COURSES:			
1.	https://www.coursera.org/specializations/improve-english			
2.	https://www.udemy.com/course/common-english-grammar-mistakes-and-how-to-fix-them-sample			
	WEB REFERENCES:			
S.No.	Publisher	Website link	Type of Content	
1.	British Council	https://learnenglish.britishcouncil.org/grammar	LSRW Skills	
2.	English Grammar	https://www.englishgrammar.org/lessons	Grammar	
	VIDEO REFERENCES:			
S.No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	YouTube	Sam & Ash	English Grammar	https://www.youtube.com/watch?v=aOsILFNgtIo
2.	Oxford Class	Kasia & Michael	Model Verbs	https://www.oxfordonlineenglish.com/free-english-grammar-lessons

Mapping of COs with POs and PSOs														
COs	POs											PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1								1	3		1			
CO2								1	3		1			
CO3								1	3		1			
CO4								1	3		1			
CO5								1	3		1			
Average								1.0	3.0		1.0			
1-Low, 2 -Medium, 3-High.														


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BE23MA201	CALCULUS FOR ENGINEERS	CP	L	T	P	C
		3	2	1	0	3
Programme & Branch	COMMON TO ALL B.E./B.TECH. BRANCHES	Version: 1.1				
Course Objectives:						
1.	To understand the concepts of matrices for analyzing physical phenomena involving continuous change.					
2.	To identify the concepts of differential calculus and various techniques.					
3.	To explain various techniques in solving ordinary differential equations.					
4.	To understand the methodologies involved in solving problems related to fundamental principles of integral calculus.					
5.	To apply the knowledge of functions of several variables.					
	INTRODUCTION (Not for Examination)					2
Importance	Importance: Calculus plays a crucial role in engineering by providing the tools needed to understand and model changes in systems, which is the fundamental to solve many engineering problems. In modern fields like data science and AI, engineers use calculus for machine learning algorithms, predictive modelling, and real-time system performance evaluation.					
Real-life Example(s)	Population Dynamics, Area and Volume calculation, Image Processing.					
Linkages	-					
UNIT-I	MATRICES					6+3
	Essential of matrices - Eigenvalues and Eigenvectors of a real matrix- Properties of Eigenvalues and Eigenvectors (Excluding proof)- Problems- Statement and application of Cayley - Hamilton theorem (Excluding proof)- Problems- Reduction of a quadratic form to canonical form by orthogonal transformation- Nature of quadratic forms - Engineering Applications.					
UNIT-II	DIFFERENTIAL CALCULUS					6+3
	Differentiation an outline - Limit of a function - Continuity - Derivatives- Differentiation rules - Maxima and Minima of functions of one variable - Engineering Applications.					
UNIT-III	ORDINARY DIFFERENTIAL EQUATIONS					6+3
	A View on ODE's - Second and Higher order linear differential equations with constant coefficients - Method of variation of parameters- Homogeneous equation of Cauchy's and Legendre's type - Engineering Applications.					

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UNIT – IV	INTEGRAL CALCULUS	6+3
	Essential of Integration - Definite and Indefinite Integrals - Substitution rule - Integration by parts - Multiple integral - simple problems- Area enclosed by plane curves- Engineering Applications.	
UNIT–V	FUNCTIONS OF SEVERAL VARIABLES	6+3
	Introduction to PDEs- Classification of PDE's (Elliptic, Parabola, Hyperbola) and its Engineering Application(Laplace, Wave and Heat equations)- Homogeneous functions and Euler's theorem - Total derivatives - Jacobian's- Maxima and minima of functions of two variables- Lagrange's method of undetermined multipliers.	
	Total (LT)	47 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Apply knowledge of matrices with the concepts of eigenvalues to study their problems in core area.	L3 – Apply
CO2	Apply differential calculus tools in solving various application problems.	L3 – Apply
CO3	Solve basic application problems described by second and higher order linear differential equations with constant coefficients.	L3 – Apply
CO4	Apply basic concepts of integration to evaluate line, surface and volume integrals.	L3 – Apply
CO5	Apply the basic techniques and theorems of functions of several variables in other area of mathematics.	L3 – Apply
	TEXTBOOKS:	
1.	Kreyzig E., "Advanced Engineering Mathematics", Tenth Edition, John Wiley and sons, 2011.	
2.	T.Veerarajan "Engineering Mathematics", 5 th Edition, Tata McGraw hill Education Pvt. Ltd,2006.	
	REFERENCE BOOKS:	
1.	Grewal B.S., "Higher Engineering Mathematics", 41 st Edition, Khanna Publishers, New Delhi,2011.	
2.	Narayanan S. and Manicavachagom Pillai.T.K., "Calculus", Volume I and II, Viswanathan S, Printers & Publishers Pvt. Ltd, 2009.	

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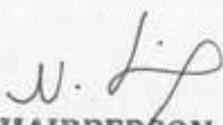
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WEB REFERENCES:				
S.No.	Publisher	Website link		Type of Content
1.	The International Linear Algebra Society	https://journals.uwyo.edu/index.php/ela/article/view/7783/6463		Journal
2.	Tutorials point	https://ncas.in/wp-content/uploads/2024/08/K.ALAMELU-PPT-OF-VECTOR-SPACE.pdf		Lecture Notes
VIDEO REFERENCES:				
S.No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	NPTEL	Prof.Jitendra kumar IIT Karagpur	LECTURE	https://youtu.be/4QFsiXfgbzM
2.	NPTEL	Dr.D.N.Panduy , IIT Roorkee	LECTURE	https://youtu.be/LompT8T-9y4

Mapping of COs with POs and PSOs														
COs	POs											PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2												
CO2	3	2												
CO3	3	2												
CO4	3	2												
CO5	3	2												
Average	3.0	2.0												

1-Low, 2 -Medium, 3-High.


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BE23PH204	ENGINEERING PHYSICS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	COMMON TO B.E. (EEE, ECE AND ECX) BRANCHES	Version: 1.1				
Course Objectives:						
1.	To introduce the electric and magnetic properties of materials and their applications.					
2.	To identify the basic concepts of semiconductors and their applications.					
3.	To elaborate fiber optics and lasers concepts.					
4.	To introduce the basics of oscillations and dielectric materials.					
5.	To outline the concepts of nano structures and devices.					
	INTRODUCTION (Not for Examination)					2
Importance	Electrical properties of materials, Electron transport mechanism, Fiber optic communication using laser source, Induction coil, Capacitors and Nanomaterials.					
Real-life Example(s)	Semiconductors, Electronics, Optical fibers (LIDAR), Oscillators, Nanotechnology.					
Linkages	Pre requisites : Newton mechanics, Electrostatics Future Courses: Quantum Computing, Nano Technology, Spintronics					
UNIT-I	ELECTRICAL AND MAGNETIC PROPERTIES OF MATERIALS					9
	Classical free electron theory - expression for electrical Conductivity- thermal conductivity, expression- Wiedemann - Franz law- Fermi - Dirac statistics - degenerate states- density of energy states- classification of magnetic material- domain theory of ferromagnetism- Quantum Interference devices.					
UNIT-II	SEMICONDUCTING MATERIALS					9
	Introduction - Energy band diagram- direct and indirect band gap semiconductors - intrinsic semiconductors (Qualitative)- extrinsic semiconductors - carrier concentration in N-type and P-type semiconductors- transport phenomena- carrier transport in semiconductor: random motion, drift, mobility and diffusion- Hall Effect and devices- Ohmic contacts.					
UNIT-III	FIBER OPTICS AND LASERS					9
	Basics of optical fibers - types of optical fibers- principle and propagation of light through optical fiber - fiber optic communication - Active and passive sensors: pressure and displacement - Basics of LASER - Einstein's coefficients - CO ₂ laser, Semiconductor laser- applications of lasers in industry.					

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UNIT - IV	OSCILLATIONS AND DIELECTRIC MATERIALS	9
	Introduction to oscillations - Simple harmonic motion - resonance- analogy between electrical and mechanical oscillating Systems - dielectric materials (piezo, pyro and ferro) - electronic and ionic polarization- dielectric loss- internal field & Clausius - Mosotti relation - dielectric breakdown.	
UNIT - V	NANO MATERIALS	9
	Basics of Nano materials- preparation, properties and applications- carbon nanotubes: properties, preparation techniques and applications- spintronic devices and applications- quantum well laser - nano materials for high voltage insulation.	
	Total	47 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. such problems can be given as Assignments and evaluated as IA only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Summarize the electric, magnetic materials and applications.	L3 - Apply
CO2	Acquire the concepts of semiconducting materials and their applications.	L3 - Apply
CO3	Rephrase the basics of fiber optics and lasers.	L2 - Understand
CO4	Summarize the basic physics of oscillations and dielectrics properties.	L2 - Understand
CO5	Describe the basics of nanomaterials, properties and applications.	L2 - Understand
	TEXTBOOKS:	
1.	D.K. Bhattacharya, Poonam Tandon, "Engineering Physics", Oxford University press, 2015.	
2.	S.O. Kasap. Principles of Electronic Materials and Devices, McGraw Hill Education (Indian Edition), 2020.	
3.	Jaspri Singh, Semiconductor Optoelectronics: Physics and Technology, McGraw-Hill Education (Indian Edition), 2019.	
	REFERENCE BOOKS:	
1.	Jaspri Singh, "Semiconductor Devices: Basic Principles", Wiley (Indian Edition), 2007.	
2.	Charles Kittel, Introduction to Solid State Physics, Wiley India Edition, 2019.	
3.	Mark Fox, Optical Properties of Solids, Oxford University Press, 2001.	

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		WEB REFERENCES:			
S.No.	Publisher	Website link		Type of Content	
1.	Brainkart	https://www.brainkart.com/subject/Physics-for-Electronics-Engineering_272/		e-resources(Electrical properties)	
2.	ScienceDirect	https://www.sciencedirect.com/topics/physics-and-astronomy/quantum-well-lasers		e-resources (nanotechnology)	
		VIDEO REFERENCES:			
S.No.	Video Details	Name of the Expert	Type of Content	Video link	
1.	NPTEL (Electron Hole Concentration)	Dr. Sapna Katiyar	Video Lecture	https://youtu.be/JfOYBuG7-Dg?feature=shared	
2.	NPTEL (Laser Technology)	Dr.Shrikrishna N.Joshi	Video Lecture	https://youtu.be/XkU9jWg49rg?si=O8v00L83vBMWKOne	

Mapping of COs with POs and PSOs														
COs	POs											PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
C01	2	1												
C02	2	1												
C03	2	1												
C04	2	1												
C05	2	1												
Average	2.0	1.0												
1-Low, 2 -Medium, 3-High.														


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BE23CY201	ENGINEERING CHEMISTRY	CP	L	T	P	C
		3	3	0	0	3
Programme& Branch	COMMON TO ALL B.E. / B.TECH. BRANCHES	Version: 1.1				
Course Objectives:						
1.	To make the students conversant with boiler feed water requirements, related problems and water treatment techniques.					
2.	To impart knowledge on the Preparation, properties and applications of engineering materials.					
3.	To understand Principles of electrochemical reactions, redox reactions in corrosion of materials and basics of polymers					
4.	To introduce the principles and generation of energy in batteries, nuclear reactors, solar cells, wind mills and fuel cells.					
5.	To aware and understand about industry safety precautions and its standards.					
	INTRODUCTION (Not for Examination)					2
Importance	For engineering students, Engineering Chemistry combines chemical principles with engineering practices to address industrial and environmental challenges. It emphasizes the design, optimization, and innovation of chemical processes and materials in industries such as energy, polymer and materials science. By applying chemistry to practical problems, students learn to develop sustainable technologies and create advanced materials for various engineering applications.					
Real-life Example(s)	Water purification process – Smart screen – Corrosion control – Storage devices					
Linkages	Pre-Requisite : Nil Future courses : Environmental Science and Sustainability					
UNIT-I	WATER AND ITS TREATMENT					9
	Need of water treatment – applications, Water resources - Hardness of water- types – expression of hardness- units – estimation of hardness of water by EDTA- numerical problems, treatment of boiler feed water- Internal treatment (phosphate, colloidal, sodium aluminate and calgon conditioning), external treatment – Ion exchange process, zeolite process – desalination of brackish water- Reverse Osmosis.					
UNIT-II	NANO MATERIALS AND PREPARATIONS					9
	Applications of nanomaterials in medicine, agriculture, energy, electronics and catalysis, Optical material for smart screen (LED, LCD & OLED), Fundamentals of nano science - Basics: Distinction between molecules, nanomaterials and bulk materials - Size- dependent properties (optical, electrical, mechanical and magnetic) - Types of nanomaterials-Definition, properties and uses of – nanoparticle, nanocluster, nanorod, nanowire and nanotube - Preparation of nanomaterials					

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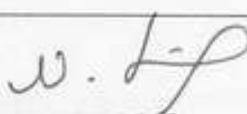
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UNIT-III	ELECTROCHEMISTRY AND POLYMERS	9
	Electro chemistry; Need and applications, Electrochemical cell- redox reaction- electrochemical series and its significance- Nernst equation, Corrosion- causes- factors- types- chemical, electrochemical corrosion (galvanic, differential aeration), corrosion control - electrochemical protection- sacrificial anode method, Polymers; Need and applications - Classification of polymers- Natural and synthetic; Thermoplastic and Thermosetting, Functionality - Degree of polymerization. Preparation, properties and uses of Nylon 6,6, and Epoxy resin.	
UNIT - IV	BATTERIES AND FUEL CELLS	9
	Batteries: Need and applications, Energy storage devices classification- Batteries - Types of batteries, Primary battery- dry cell, Secondary battery- lead acid battery - lithium ion battery - Electric vehicles introduction - working principles - Fuel cells - H ₂ -O ₂ fuel cell - Microbial fuel cell - Super capacitors - Storage principle - types and examples.	
UNIT - V	CHEMISTRY, ENVIRONMENT AND WASTE MANAGEMENT	9
	Industry safety procedures and precautions- Industry standards ISO Audits - Green chemistry - principles of green chemistry - E- wastes and its management- Carbon foot print and its calculations - CO ₂ emission and its impact on environment- Reduction of CO ₂ emission by various techniques.	
	Total (L)	47 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. such problems can be given as Assignments and evaluated as IA only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Infer the quality of water from quality parameter data and propose suitable treatment methodologies to treat water.	L2 - Understand
CO2	Identify and understand basic concepts of nanoscience and nanotechnology in designing the synthesis of nanomaterials for engineering and technology applications.	L2 - Understand
CO3	Know about basics of electro chemistry and polymers	L2 - Understand
CO4	Understand about the various advanced power storage devices working principles and its applications.	L2 - Understand
CO5	Understand the basic concepts of safety standards in industry and carbon credit.	L2 - Understand


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	TEXTBOOKS:			
1.	R.K. Jain and Prof. Sunil S. Rao Industrial Safety, Health and Environment Management Systems khanna publisher, 2000.			
2.	S. S. Dara and S. S. Umare, "A Textbook of Engineering Chemistry", S. Chand & Company LTD, New Delhi, 2015.			
3.	P. C. Jain and Monika Jain, "Engineering Chemistry" Dhanpat Rai Publishing Company (P) LTD, New Delhi, 2015.			
	REFERENCE BOOKS:			
1.	John Ridley & John Channing Safety at Work: Routledge, 7th Edition, 2008.			
2.	B. S. Murty, P. Shankar, Baldev Raj, B. B. Rath and James Murday, "Text book of nanoscience and nanotechnology", Universities Press-IIM Series in Metallurgy and Materials Science, 2018.			
3.	O.G. Palanna, "Engineering Chemistry" McGraw Hill Education (India) Private Limited, 2nd Edition, 2017.			
	WEB REFERENCES:			
S.No.	Publisher	Website link	Type of Content	
1.	Central Pollution Control board	https://cpcb.nic.in/wqstandards/	Norms and Standards	
2.	National Library of medicine	https://www.ncbi.nlm.nih.gov/books/NBK21032/	Causes of diseases	
3.	Center for Green Chemistry & Green Engineering at Yale	https://greenchemistry.yale.edu/about/principles-green-chemistry	Principles of Green Chemistry	
	VIDEO REFERENCES:			
S.No.	Video Details	Name of the Expert	Type of Content	Video link
1.	NPTEL	Prof. Bhanu Prakash Vellanki IIT Roorkee	Video Lecture	https://onlinecourses.nptel.ac.in/noc22_ce55/preview
2.	NPTEL	Prof. Ashok IIT Madras	Video Lecture	https://www.youtube.com/watch?v=A3fHQsIkYeU

Mapping of COs with POs and PSOs														
COs	POs											PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	1									1			
CO2	2			1		2								
CO3	3	1	2	1		2					2			
CO4	3	2	2	1		1					1			
CO5	3	1	2	1		2					2			
Average	2.8	1.3	2	1		1.8					1.5			

1-Low, 2 -Medium, 3-High.

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BE23GE301	OVERVIEW OF ENGINEERING AND TECHNOLOGY	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	COMMON TO ALL B.E. / B.TECH. BRANCHES	Version: 1.1				
Course Objectives:						
1.	To outline the basics of the Civil Engineering Program					
2.	To learn the fundamentals of Mechanical Engineering					
3.	To impart knowledge on fundamental concepts and recent trends in the field of Electrical and Control Systems					
4.	To provide the overview of the Electronics and Communication Engineering Program					
5.	To provide a comprehensive overview of the field of Computer Science, from the historical roots to most cutting-edge developments					
UNIT I	INTRODUCTION TO ENGINEERING AND TECHNOLOGY (NOT FOR EXAMINATION)					7
	Science, Engineering and Technology(E&T), Approaches for a Scientific process vs an Engineering process; Engineering Product Life Cycle, processes in Engineering Design Methodology with few examples; various branches in Engineering and Technology (Traditional and Recent), Impact of E&T on human life, (pros & cons); Activities performed by an Engineer, Interdisciplinary nature of real world problems; Revised Bloom's Taxonomy Levels (BTL) and Engineering Teaching Learning Process (TLP); Structure, Duration and BTL levels in UG, PG & Ph.D. level Education in E&T, History of E&T development and emerging fields in E&T.					
UNIT II	OVERVIEW OF CIVIL ENGINEERING					6
	Introduction— Major Areas of Study: Architecture and Town Planning, Structural Engineering, Construction Engineering and Management, Hydrology and Water Resources Engineering, Environmental Engineering, Transportation Engineering — Historical Perspective— Few Practical Applications*: Single Story Residential Building, Roads and Highway Network, Dam, Canals and Irrigation layout, Sewage System and its Treatment — Recent Developments / Current Areas of Research.					
UNIT III	OVERVIEW OF MECHANICAL ENGINEERING					8
	Introduction— Major Areas of Study: World Energy Scenario, CO2 and other Emissions and Climatic Change, Energy Conservation Systems, Mechanical Design, Manufacturing and Industrial Engineering — Historical Perspective— Few Practical Applications* : Thermal Power Plant, Air Conditioning Systems, Automobile (Car / Truck), Mechanical Design of a Component using CAD, Assembly Line of a Car manufacturing Plant Machines in a Textile Spinning Industry — Recent Developments / Current Areas of Research.					

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UNIT IV	OVERVIEW OF ELECTRICAL, ELECTRONICS AND CONTROL SYSTEMS ENGINEERING	9
	Introduction— Major Areas of Study: Electrical Power Generation, Transmissions and Distributions, Motors, Lighting Systems, Electrical Appliances, Sensors, Instrumentation and Control Systems, Distributed Multimode Power Generation and Distributed Power Consumption — Historical Perspective - Few Practical Applications*: Generators, Transmission Systems, Home Appliances, Rating, Load Estimations, Wiring, Electrical Appliances (Induction Stove, BLD C Fan vs Ordinary Fan). Introduction Control Systems Layout, Open Loop and Closed Loop, System Responsive or Time Constant — Few Practical Applications*: Various types of Control Systems: Mechanical, Pneumatic, Electrical, Electronic (Microprocessor based), Embedded Control Systems, PLCs, SCADA, Computer Based Control Systems.	
UNIT V	OVERVIEW OF ELECTRONICS AND COMMUNICATION ENGINEERING	9
	Introduction, Major Areas of Study: Electronic Devices and Circuits, Analog Electronics, Digital Electronics, Embedded Systems, Integrated Circuits & VLSI — Historical Perspective— Fee Practical Applications*: Audio Systems, Washing Machine, Automotive Electronic Systems — Recent Developments / Current Areas of Research, Introduction — Major Areas of Study: Signal Processing, Analog and Digital Communication, Data Communications and Networking — Historical Perspective— Few Practical Applications: Text to Speech / Voice to Text Application in Google Search, Wired Communications Network, Wireless Communications Network, Satellite Communications, IoT Communications Network - Recent Developments / Current Areas of Research.	
UNIT VI	OVERVIEW OF COMPUTER SCIENCE AND ENGINEERING	6
	Introduction: Evolution of Computers / Generation Computers — Major Areas of Study: Computer Hardware, Programming Languages, Operating Systems, Application Software, Database Management Systems (DBMS), Computer Networks, Internet and Computer Security, Web Technology, Social Media, Mobile Application- Recent Developments / Current Areas of Research: Artificial Intelligence AI and Machine Learning (ML), Internet of Things IoT), Block Chain, Big Data Analytics, Cyber Security, Cloud Computing.	
	* Purpose or Use, Actual System (Photo), Layout or Block Diagram, Description, Operational Aspects and Inputs/Outputs are to be taught (Descriptive level only).	
TOTAL (L)		45 Periods

w.e.f. 13/10/2025

Passed in BoS Meeting held on 09/05/2025

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	OPEN ENDED PROBLEMS/QUESTIONS	
	Course specific Open-Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment (IA) only and not for the End semester Examinations.	

	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Identify the major areas and relate their current trends in Civil Engineering.	L2- Understand
CO2	Explain the principles behind various mechanical systems and components.	L2- Understand
CO3	Identify different Electricals and Control Systems applied in the Engineering field.	L2- Understand
CO4	Relate the various Electronics and Communication Engineering Systems involved in real life.	L2- Understand
CO5	Understand the components of Computer Hardware, Software, and Operating Systems and their applications in real life.	L2- Understand

TEXTBOOKS:

1.	"Overview of Engineering and Technology", Lecture Notes from KIOT, 2023.
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REFERENCE BOOKS:

1.	Banapurmath N.R., & Yalliwal V.S., "Basics of Mechanical Engineering", Vikas Publishing House, 2021.
2.	G Shanmugam, M S Palanichamy, "Basic Civil and Mechanical Engineering", McGraw Hill Education; 1 st Edition, 2018.
3.	Kothari DP and I.J Nagrah, "Basic Electrical Engineering", 4 th Edition, McGraw Hill Education, 2019.
4.	Albert Malvino and David 1. Bates, "Electronic Principles (SIE)", 7 th Edition, McGraw Hill 2017.
5.	Reema Thareja, "Fundamentals of Computer", Oxford University Press, 2016.

Mapping of COs with POs and PSOs

COs	POs											PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3													
CO2	3													
CO3	3													
CO4	3													
Average	3.0													

1-Low, 2 -Medium, 3-High

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BE23MC901	தமிழர் மரபு / HERITAGE OF TAMILS	CP	L	T	P	C
		1	1	0	0	1
Programme & Branch	COMMON TO ALL B.E./B.TECH. BRANCHES	Version: 1.1				
Students can write the examination either in Tamil or in English						
Course Objectives:						
1.	To Learn about the Indian language family and Tamil literature.					
2.	To acquire knowledge on the history of rock paintings and modern paintings.					
3.	To Learn about the arts and games of Tamils.					
4.	To learn knowledge on Thinaï Theory in Tolkappiyam and Sanga Literature.					
5.	To learn the national consciousness of Tamils and Tamil culture.					
	INTRODUCTION (Not for Examination)					1
Importance	Tamil heritage is the specialty of the Tamil language. It includes the Dravidian language family and Tamil literature. Tamil arts. Tamil games. Tolkappiyam. Tamil lifestyle through Sangam literature Tamil national consciousness.					
UNIT-I	LANGUAGE AND LITERATURE					3
	Language Families in India - Dravidian Languages- Tamil as a Classical Language - Classical Literature in Tamil- Secular Nature of Sangam Literature- Distributive Justice in Sangam Literature - Management Principles in Thirukural - Tamil Epics and Impact of Buddhism & Jainism in Tamil Land - Bakthi Literature Azhwars and Nayanmars - Forms of minor Poetry - Development of Modern literature in Tamil - Contribution of Bharathiyar and Bharathidhasan.					
UNIT-II	HERITAGE - ROCK ART PAINTINGS TO MODERN ART-SCULPTURE					3
	Hero stone to modern sculpture - Bronze icons - Tribes and their handicrafts - Art of temple car making - Massive Terracotta sculptures, Village deities, Thiruvalluvar Statue at Kanyakumari, Making of musical instruments - Mridhangam, Parai, Veenai, Yazh and Nadhaswaram - Role of Temples in Social and Economic Life of Tamils.					
UNIT-III	FOLK AND MARTIAL ARTS					3
	Therukoothu, Karagattam, Villu Pattu, Kaniyan Koothu, Oyillattam, Leatherpuppetry, Silambattam, Valari, Tiger dance - Sports and Games of Tamils.					

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UNIT – IV	THINAI CONCEPT OF TAMILS	3	
	Flora and Fauna of Tamils & Aham and Puram Concept from Tholkappiyam and Sangam Literature - Aram Concept of Tamils - Education and Literacy during Sangam Age - Ancient Cities and Ports of Sangam Age - Export and Import during Sangam Age - Overseas Conquest of Cholas.		
UNIT-V	CONTRIBUTION OF TAMILS TO INDIAN NATIONAL MOVEMENT AND INDIAN CULTURE	3	
	Contribution of Tamils to Indian Freedom Struggle - The Cultural Influence of Tamils over the other parts of India - Self-Respect Movement - Role of Siddha Medicine in Indigenous Systems of Medicine- Inscriptions & Manuscripts- Print History of Tamil Books.		
	Total	16 Periods	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy	
CO1	Find the Indian language family and Tamil literature.	L1 - Remember	
CO2	Explain the evolution of contemporary and rock painting arts.	L1 - Remember	
CO3	Describe the games and arts in Tamils.	L1 - Remember	
CO4	Interpret the Thinaï theories in Tholkappiyam and Sanga literature.	L2 - Understand	
CO5	State the need of national consciousness of Tamils and Tamil culture.	L1 - Remember	
	TEXTBOOKS:		
1.	டாக்டர் கே.கே. பிள்ளை, "தமிழக வரலாறு மக்களும் பண்பாடும்", (வெளியீடு, தமிழ்நாடு பாடநூல் கல்வியியல் பணிகள் கழகம்), 2021.		
2.	முனைவர் இல. சுந்தரம், "கணினித்தமிழ்", (விகடன் பிரசுரம்), 2015.		
	REFERENCE BOOKS:		
1.	"கீழடி - வைகை நதிக்கரையில் சங்ககால நகர நாகரிகம்", (தொல்லியல் துறை வெளியீடு).		
2.	"பொருதை - ஆற்றங்கரை நாகரிகம்", (தொல்லியல் துறை வெளியீடு), 2021.		
	WEB REFERENCES:		
S.No.	Publisher	Website link	Type of Content
1.	wikipedia	http://www.news.mowval.in/News/tamilnadu/Nano-9202.html	e-resources
2.	wikipedia	https://ta.wikipedia.org/wiki	e-resources

w.e.f. 13/10/2025

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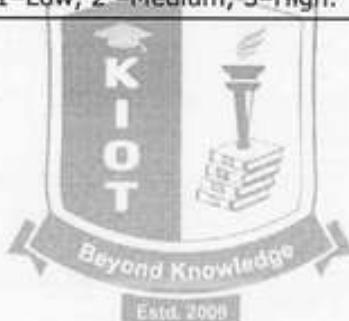
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VIDEO REFERENCES:				
S.No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	YouTube	Dr.B.Anandhi Meena	Lectures	https://www.youtube.com/watch?v=x0Sy7QVDJI4
2.	YouTube	Dr.S.Prabavathi	Lectures	https://youtu.be/AXrQI7gdPPc?si=qEsphCi-xZquin4B

Mapping of COs with POs and PSOs														
COs	POs											PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1									2		3			
CO2											2			
CO3							1		2		3			
CO4							1		1		1			
CO5							1		1		3			
Average							1.0		1.5		2.4			
1-Low, 2 -Medium, 3=High.														




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BE23GE306	PROBLEM SOLVING AND C PROGRAMMING	CP	L	T	P	C
		5	3	0	2	4
Programme & Branch	Common to B.E. (CIVIL, ECE, ECX, EEE and MECH) Branches	Version: 1.1				
Course Objectives:						
1.	To learn how to think algorithmically to solve a problem.					
2.	To gain knowledge of fundamental programming concepts in C language.					
3.	To explore the basic concept of various control flow statements and arrays.					
4.	To learn pointers and modular programming principles.					
5.	To gain proficiency in structures and unions.					
	INTRODUCTION (Not for Examination)					2
Importance	- Problem solving helps to develop logical thinking, analytical ability, and a systematic approach to tackling challenges. - C programming builds a strong base in programming concepts and teaches how software interacts closely with hardware. It is the foundation for many modern languages. - Learning problem-solving helps to design solutions for real-world tasks, and C programming implements them efficiently.					
Real-life Example(s)	Route Planning, Banking Applications, Scheduling, Operating Systems, Embedded Systems					
Linkages	Future courses: Object Oriented Programming using C++ and Java, Data Structures and SQL, Microcontrollers and Embedded Systems.					
UNIT-I	COMPUTATIONAL THINKING					9
	Computational Thinking: Overview - Key Techniques - Overview of Software Development Life Cycle - Algorithmic Thinking: Introduction - Elements: Sequence - Selection and Repetition - Representation: Flow Chart - Overview of Flowgorithm Tool - Pseudo-code - Programs - Introduction to programming languages.					
UNIT-II	BASICS OF C PROGRAMMING					9
	Introduction: Features - Structure of C Programming - Compiling - Executing and Debugging - Character Set - Tokens: Keywords - Identifiers - Constants - Strings - Operators - Special Symbols - Data Types Expression - Precedence and Associativity - Evaluating Expression - Type Conversion - Input and Output: Unformatted Input and Output - Formatted Input and Output.					
UNIT-III	CONTROL FLOW STATEMENTS AND ARRAYS					9
	Control Flow Statements: Sequence - Selection - Looping - Jumping Statements. Arrays: Introduction - Declaration and Initialization of Single Dimensional Arrays - Array Operations - Declaration and Initialization of Two-Dimensional Arrays - Character					

w.e.f. 27/06/2025

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	Arrays (Strings): Declaring and Initializing Strings - Reading and Writing Strings - String Operations.	
UNIT-IV	POINTERS AND FUNCTIONS	9
	Pointers: Introduction to Pointers - Pointer operators - Pointer arithmetic - Arrays and pointers - Array of pointers. Function: Need of Function - Elements - Types - Parameter passing: Pass by value - Pass by reference - Recursion - Storage Classes.	
UNIT-V	STRUCTURES, UNIONS AND BIT FIELDS	9
	Structures: Introduction - Declaring and Defining Structure Variables - Accessing Structure Members - Structure Initialization - Nested structures - Array of structure - typedef - Union - Bitfields.	
	Total (L)	47 Periods
	LIST OF EXPERIMENTS:	
1.	Implementation of algorithms, flowcharts and pseudo codes for simple problems.	
2.	Implementation of programs using basic programming constructs.	
3.	Implementation of if, if-else, nested if and switch statements.	
4.	Implementation of while, do-while, for loops.	
5.	Implementation of one dimensional array and two dimensional array.	
6.	Implementation of programs using strings.	
7.	Implementation of pointer concept.	
8.	Implementation of function calls, call by value and reference, recursion.	
9.	Implementation of structures and nested structures.	
10.	Implementation of array of structures.	
	Total (P)	30 Periods
	Total (L+P)	77 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	

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	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy		
CO1	Construct algorithmic solutions for a given computational problem.	L3 - Apply		
CO2	Demonstrate the understanding of fundamental concepts of C programming.	L3 - Apply		
CO3	Utilize appropriate control flow statements and arrays to solve programming problems effectively.	L3 - Apply		
CO4	Develop programs using pointers and modular programming principles.	L3 - Apply		
CO5	Implement various concepts of structures and unions.	L3 - Apply		
TEXTBOOKS:				
1.	Reema Thareja, "Programming in C", 2 nd Edition, Oxford University Press, 2016.			
2.	E Balagurusamy, "Programming in ANSI C", 8th Edition, McGraw Hill Education (India) Private Ltd., 2019.			
3.	Yashavant Kanetkar, "Let us C: Authentic Guide to C Programming Language", 17th Edition, BPB Publications, 2020.			
REFERENCE BOOKS:				
1.	Byron S Gottfried and Jitendar Kumar Chhabra, "Programming with C", 4 th Edition, McGraw Hill Education (India) Private Ltd., 2019.			
2.	Pradip Dey and Manas Ghosh, "Programming in C", 2 nd Edition, Oxford University Press, 2011.			
3.	Brian W. Kernighan and Dennis M. Ritchie, "The C Programming language", 2 nd Edition, Pearson Education India, 2015.			
WEB REFERENCES:				
S.No	Publisher	Website link	Type of Content	
1.	Geeksforgeeks	https://www.geeksforgeeks.org/c-programming-language/	Web Content	
2.	TutorialsPont	https://www.tutorialspoint.com/cprogramming/index.htm	Web Content	
VIDEO REFERENCES:				
S.No	Video Details	Name of the Expert	Type of Content	Video Link
1.	YouTube	Bro Code	Lecture and demonstration	https://youtube.com/playlist?list=PLZPZq0r_RZOOzY_vR4zJM32SqsSInGMwe
2.	YouTube	Navin Reddy	Lecture and demonstration	https://youtube.com/playlist?list=PLsyebzWxI7oBxHp43xQTFrw9f1CDPR6C
3.	YouTube	Padma Manandhar	Lecture and demonstration	https://youtube.com/playlist?list=PL98qAXLA6aftD9ZInjp

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Mapping of COs with POs and PSOs														
COs	POs											PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	2	1										
CO2	3	2	2	1										
CO3	3	2	2	1										
CO4	3	2	2	1										
CO5	3	2	2	1										
Average	3.0	2.0	2.0	1.0										
1-Low, 2 -Medium, 3-High.														


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BE23BS201	PHYSICS AND CHEMISTRY LABORATORY	CP	L	T	P	C
		4	0	0	4	2
Programme & Branch	COMMON TO ALL B.E. / B.TECH. BRANCHES	Version: 1.1				
PHYSICS LABORATORY						
Course Objectives:						
1.	To learn the proper use of various kinds of physics laboratory equipment.					
2.	To learn problem solving skills related to physics principles and interpretation of experimental data.					
3.	To determine error in experimental measurements and techniques used to minimize such error.					
4.	To make the student as an active participant in each part of all lab exercises.					
	List of Experiments.					
1.	Torsional pendulum - Determination of rigidity modulus of wire and moment of inertia of regular and irregular objects.					
2.	Uniform bending – Determination of Young’s modulus.					
3.	Non-uniform bending - Determination of Young’s modulus.					
4.	Air wedge - Determination of thickness of a thin sheet/wire.					
5.	Optical fibre -Determination of Numerical Aperture and acceptance angle Compact disc- Determination of width of the groove using laser.					
6.	Determination of band gap of semiconductors.					
7.	LASER – Determination of the wavelength of the LASER using grating.					
8.	** Study experiment on application of physics in a real time problem– 1					
9.	** Study experiment on application of physics in a real time problem– 2					
10.	** Study experiment on application of physics in a real time problem– 3					
	**Experiential learning: Study on real time problem– 1,2,3					
	Total					30 Periods
	** ONLY FOR INTERNAL EVALUATION					
	Course Outcomes: Upon completion of this course, the students will be able to:					BLOOM’S Taxonomy
CO1	Understand the functioning of various physics laboratory equipment.					L3 - Apply
CO2	Use graphical models to analyze laboratory data.					L3 - Apply

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CO3	Use mathematical models as a medium for quantitative reasoning and describing physical reality.	L3 - Apply
CO4	Access, process and analyze scientific information.	L3 - Apply
CO5	Solve problems individually and collaboratively.	L3 - Apply

Mapping of COs with POs and PSOs														
COs	POs											PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2												
CO2	3	1												
CO3	3	2												
CO4	2	1												
CO5	2	1												
Average	2.0	2.0												
1-Low, 2 -Medium, 3-High.														

CHEMISTRY LABORATORY	
Course Objectives:	
1.	To inculcate experimental skills to test basic understanding of water quality parameters, such as, acidity, alkalinity, hardness, DO, chloride and copper.
2.	To induce the students to familiarize with electroanalytical techniques such as, pH metry and conductometry in the determination of impurities in aqueous solutions.
3.	To demonstrate the analysis of metal ions by analytical techniques.
List of Experiments.	
1.	Estimation of alkalinity in water sample using Na ₂ CO ₃ as primary standard.
2.	Determination of total, temporary & permanent hardness of water by EDTA method.
3.	Determination of Dissolved oxygen content of water sample by Winkler's method.
4.	Determination of chloride content of water sample by argentometric method.
5.	Determination of strength of given hydrochloric acid using pH meter.
6.	Determination of strength of acids in a mixture of acids using conductivity meter.

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7.	Conductometric titration of barium chloride against sodium sulphate (precipitation titration)	
8.	** Study experiment on application of chemistry in a real time problem- 1	
9.	** Study experiment on application of chemistry in a real time problem- 2	
10.	** Study experiment on application of chemistry in a real time problem- 3	
	**Experiential learning : Study on real time problem- 1,2,3	
	Total	30 Periods
	** ONLY FOR INTERNAL EVALUATION	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Identify the quality of water samples with respect to their acidity, alkalinity, hardness and dissolved oxygen.	L3 - Apply
CO2	Determine the amount of metal ions through volumetric analysis.	L3 - Apply
CO3	Use the graphical models to analyze laboratory data.	L3 - Apply
CO4	Equipped with basic knowledge on conductivity meter for measurement of conductance of water sample.	L3 - Apply
CO5	Make use of the electroanalytical techniques to identify the impurities in solution.	L3 - Apply

Mapping of COs with POs and PSOs														
COs	POs											PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3		1			2					2			3
CO2	3	1	2			1.5					1			3
CO3	3	12	1	1		0.5								3
CO4	2	1	2			2								2
CO5	2	1	2		1	2					1			2
Average	2.6	1.3	1.6	1.0	1.0	1.4					1.3			2.6
1-Low, 2 -Medium, 3-High.														

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BE23GE305	ENGINEERING PRACTICES LABORATORY	CP	L	T	P	C
		4	0	0	4	2
Programme & Branch	COMMON TO ALL B.E. / B.TECH. BRANCHES	Version: 1.1				
Course Objectives:						
1.	To practice welding, sheet metal and machine assembly.					
2.	To practice basic building plan, pipelining and woodwork.					
3.	To practice electric wiring and precautions for household applications and Power generation.					
4.	To practice soldering and develop electronic devices for household applications.					
LIST OF EXPERIMENTS/EXERCISES:						
GROUP – A (MECHANICAL& CIVIL)						
	MECHANICAL ENGINEERING PRACTICES	15				
MODULE 1	HANDS-ON EXPERIMENT					
1.	Make a Steel Chair using Welding Technique.					
2.	Make a Plain turning and Facing using Lathe.					
3.	Make a given component using sheet metal.					
MODULE 2	Case Study Experiments (IDENTIFICATION OF PARTS, FUNCTIONS OF EACH COMPONENT, INTEGRATION AND OVERALL WORKING)					
1.	Study of Thermal Power Plant (Steam Boiler) or Air-conditioning systems.					
2.	Study of Various Machines & Machining Processes.					
3.	Study of an Automobile –Two Wheeler/Car.					
	CIVIL ENGINEERING PRACTICES					
MODULE 1	HANDS-ON EXPERIMENT					
1.	Construct a water flow pipelining network for a residential building.					
2.	Fabricate a given truss using wooden planks.					
3.	Construct a residential building as per given building drawing using mount board/Thermocol sheet.					
MODULE 2	STUDY EXPERIMENTS					
1.	Study of an Approved building plan and various details.					
2.	Study of a Highway network and various elements.					
3.	Study of construction materials and its usage in building construction.					

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GROUP – B (ELECTRICAL& ELECTRONICS)		
	ELECTRICAL ENGINEERING PRACTICES	15
MODULE 1	HANDS-ON EXPERIMENT	
1.	House Wiring (3-pin socket, staircase wiring, Lamp load, MCB, Energy meter, fuse)	
2.	Series and Parallel Connection of UPS Batteries and Solar Panel.	
3.	Assembly of water level indicator using Arduino.	
MODULE 2	STUDY EXPERIMENTS	
1.	Study of Solar Power Generation.	
2.	Study of 22kV/440V Step-down Transformer at Powerhouse.	
3.	Study of Electrical Household Appliances (Washing Machine, Electric Kettle, Induction Stove (anyone))	
	ELECTRONICS ENGINEERING PRACTICES	15
MODULE 1	HANDS-ON EXPERIMENT	
1.	LED brightness changing based on ambient light.	
2.	Digital thermometer with LCD Display.	
3.	Voltage regulator for domestic applications.	
MODULE 2	STUDY EXPERIMENTS	
1.	Study of Audio system.	
2.	Study of AM and FM Transceiver.	
3.	Study LED TV.	
4.	Study of overall Information and Communication Technology (ICT) functional structure of KIOT (Internet Infrastructure).	
TOTAL (P)		60 Periods
Course Outcomes:		BLOOM'S Taxonomy
Upon completion of this course the students will be able to:		
CO1	Perform basic welding and sheet metal.	L3 -APPLY
CO2	Perform basic building plan, pipelining and woodwork.	L3 -APPLY
CO3	Perform electric wiring and precautions for household applications.	L3 -APPLY
CO4	Understand electronic components, signal generation, and assembly of LED TV and Audio Systems.	L3 -APPLY
REFERENCE/LAB MANUAL/SOFTWARE:		
1.	Dr.V.Ramesh babu "Engineering Practices Laboratory Manual", VRB Publisher Pvt. Ltd., Chennai, 11 th edition, 2020.	

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2.	Ramesh Singh "Applied Welding: Process, Codes and Standards", Elsevier material, First edition 2012
3.	Michael A Joyce, Ray Holder "Residential Construction Academy: Plumbing", Residential construction Academy USA.

VIDEO REFERENCES:

1.	https://www.youtube.com/watch?v=nGfVTNfNwnk
2.	https://www.youtube.com/watch?v=aJp2g1BKXVc&list=PLX2gX-ftPVXU59ggWS3t0sThVF18h5ME2

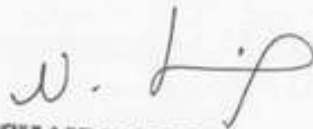
WEB REFERENCES:

1.	https://nptel.ac.in/courses/112106286
2.	https://www.brainkart.com/article/Dynamics-of-Particles_6788/

ONLINE COURSES:

1.	https://nptel.ac.in/courses/112106286
2.	https://in.coursera.org/learn/engineering-mechanics-statics

Mapping of COs with POs and PSOs														
COs	POs											PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	1			2	2		2	2					
CO2	2	1			2			2	2					
CO3	2	1			2			2	2					
CO4	2	1			2			2	2					
Average	2.0	1.0			2.0			2.0	2.0					
1-Low, 2 -Medium, 3-High														


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BE23PT801	HUMAN EXCELLENCE AND VALUE EDUCATION - I	CP	L	T	P	C
		2	1	0	1	NC
Programme & Branch	COMMON TO ALL B.E. / B.TECH. BRANCHES	Version: 1.1				
Course Objectives:						
1.	To understand oneself and manage own emotions					
2.	To learn the essence of goal-setting and time-management techniques					
3.	To learn stress management techniques for self and professional development					
4.	To inculcate the Grooming and mannerism					
5.	To acquire knowledge on social media for professional development					
	INTRODUCTION (Not for Examination)					2
Importance	Overall personality development is essential for engineers because technical knowledge alone is not enough in today's competitive and fast-changing world. Engineers need to be well-rounded professionals who can apply knowledge, communicate ideas, work in teams, and lead effectively. Ultimately, it helps future engineers lead purposeful lives and contribute positively to society.					
Real-life Example(s)	(i) A project manager or team lead sets clear goals along with timelines to ensure the efficient execution and timely completion of the project. (ii) Stress management is essential not only for maintaining productivity in professional life but also for achieving balance and well-being in personal life. (iii) Proper grooming and professional conduct enable engineers to create a positive impression and progress effectively in their career ladder. (iv) Professionals strategically use social media platforms to enhance their network, showcase their expertise, and create opportunities for career advancement.					
Linkages	Human Excellence and Value Education – II & Human Excellence and Value Education - III					
UNIT-I	SELF-AWARENESS – SELF-MOTIVATION & CONFIDENCE					3+3
	Concepts: Defining Success - Importance of Route maps to achieve Success - Understanding Need vs Want (Biological & Emotional) - Maslow's Need Theory- Emotional Intelligence - Best Practices to improve 5 Realms of EI: Self-Awareness, Self-Regulation, Self-Motivation, Empathy and Social Skills -Psychometric assessment Personality Types- Pros and Cons - Action Plan. Activity: Psychometric Test for Assessing the Personality					

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UNIT-II	GOAL SETTING AND TIME MANAGEMENT	3+3
	Concepts: Defining a Goal - Understanding Possibility and Feasibility Factors - Setting an Achievable Goal - Understanding the Differences between Micro, Small, Mid and Long Term Goals- Decision Making - Time Inventory - Time Wasters - Prioritization using UI Matrix. Activity : Preparing Short term and Long Term Goals	
UNIT-III	STRESS MANAGEMENT	3+3
	Different types of Stress - Positive vs Negative Stress - Impacts of Stress - Situation Handling - Anxiety & Adversity Management - Best Practices for Stress Management - Food for Stress Management. Activity: Role-play Challenge	
UNIT - IV	GROOMING & MANNERS	3+3
	Concepts: Importance of Grooming and Manners for Image Management - Corporate Expectations - Grooming and Manners for achievements - Etiquettes: Social, Business, Dining, Telephone, Dress, People Transaction and Road - Personal Hygiene - Cultural Adaptability. Activities: Practicing and Demonstrating various Etiquettes.	
UNIT - V	SOCIAL MEDIA	3+3
	Concepts: Understanding the Utility- Vulnerability- What(s) of Social Media- Using and Creating Contents in Blogs, Social Media Platforms, Websites - LinkedIn Profile - AI Tools - Chat GPT - Social Media for Professional Development - Do's and Don'ts in Social Media. Activity: Developing a blog, Creating LinkedIn Profile, Practice in AI tools, Developing a webpage	
	Total	32 Periods
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Be confident and motivated to plan the activities according to personality types	L2 - Understand
CO2	Set their short-term and long-term goals and manage their time effectively.	L2 - Understand
CO3	Practice stress management techniques in their personal life and career.	L2 - Understand
CO4	Practice manners and etiquettes in day-to-day life.	L2 - Understand
CO5	Use social media for professional development.	L2 - Understand
	TEXTBOOKS:	
1.	Trainer and Faculty Lecture Notes and PPT from KIOT, 2023.	

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	REFERENCE BOOKS:			
1.	Suresh Kumar E, Sreehari P, Savithri J, "Communication Skills and Soft Skills", Pearson India Education Services, 2011.			
2.	Alex K, "Soft Skills Know yourself and know the world", S. Chand & Company Pvt Ltd., 2014.			
3.	Shiv Khera, "You Can Win A Step-by-Step Tool for Top Achievers", Bloomsbury Publishing, 2013.			
4.	Norman Vincent Peale, "The Power of Positive Thinking", RHUK, 2016.			
5.	Social Media Marketing Liana Li Evans, Pearson India Education Services, 2011			
6.	Brian Tracy, "Goals", Collins, 2020			
7.	Brian Tracy, "Time Management", Amacom, 2019			
8.	Kathryn Critchley, "Stress Management Skills Training Course", Universe of Learning Ltd., 2010			
VIDEO REFERENCES:				
S.No.	Publisher	Website link	Type of Content	
1.	Skills You Need	https://www.skillsyouneed.com/ps/personal-development.html	Skills for Personal Development	
2.	Skills You Need	https://www.skillsyouneed.com/ps/personal-development.html	Skills for Personal Development	
3.	Jobscan / Hogan Short	https://www.jobscan.co/blog/5-interpersonal-skills-you-need-on-your-resume/#What-are-%20interpersonal-skills?	Interpersonal Skills	
WEB REFERENCES:				
S.No.	Video Details	Name of the Expert	Type of Content	Video link
1.	TED	John Doerr	Lecture	https://www.youtube.com/watch?v=L4N1q4RNI9I
2.	TEDx Tucson	Stephen Duneier	Lecture	https://www.youtube.com/watch?v=TQMbvJNRpLE
3.	Motivation2Study	Jordan Peterson	Lecture	https://www.youtube.com/watch?v=wsNzAuYDgy0
4.	TED-Ed	Amy Cuddy	Lecture	https://www.youtube.com/watch?v=RWZluriQUzE
ONLINE COURSES:				
1.	NPTEL Course on Enhancing Soft Skills and Personality - https://nptel.ac.in/courses/109104115			
2.	NPTEL course on Soft skills - https://nptel.ac.in/courses/109107121			

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Mapping of COs with POs and PSOs														
COs	POs											PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1								2						
CO2										2	3			
CO3								2						
CO4							2	1	2					
CO5						2	2		2					
Average						2.0	2.0	1.7	2.0	2.0	3.0			
1-Low, 2 -Medium, 3-High.														



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BE23EN102	COMMUNICATIVE ENGLISH - II	CP	L	T	P	C
		2	1	1	0	2
Programme & Branch	COMMON TO ALL B.E. / B.TECH. BRANCHES	Version: 1.1				
Course Objectives:						
1.	To enable learners to improve their language competency.					
2.	To help learners comprehend documents in a professional context.					
3.	To develop learners' skills in a professional framework.					
4.	To strengthen learners' public speaking skills.					
5.	To improve the interpersonal skills of the learners.					
	INTRODUCTION (Not for Examination)					2
Importance	• This course provides a comprehensive framework to enhance learners' proficiency in English by focusing on grammar, reading, writing and speaking. • It emphasizes precise language usage, comprehension, structured writing, effective oral communication and practical survey. • Learners are suggested to apply these skills effectively in academic, professional, and social contexts.					
Real-life Example(s)	Writing letters – Interpretation of charts – Research Proposals					
Linkages	Communicative English - I					
UNIT-I	FUNCTIONAL GRAMMAR					3+3
	Concept: Usage of Prepositions - Degrees of Comparison - Subject-verb Agreement - IfConditional Clause - Reported Speech - Common errors in English usage . Activity: Practice using worksheets.					
UNIT-II	READING FOR INFORMATION					3+3
	Concept: Comprehending a passage - Identifying a topic sentence - find specific information and prepare notes - classify the information - reading texts, essays and summarizing, ReadingTechnical Articles - Company Profile . Activity: Reading daily news - Reading comprehension.					
UNIT-III	EXTENDED WRITING					3+3
	Concept: Interpretation of charts – Pie chart, Bar chart, Flow chart - Dialogue Writing (- Writingresearch article - Project proposal - Official letters: Joining report, Placing order, Letter seeking clarification , Acknowledging prompt/quality service . Activity: letters of inviting guest - accepting / declining offer.					

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UNIT – IV	FOCUS ON SPEAKING SKILL	3+3
	Concept: Conversation Practice in real life situations - Describing process - Pronunciation practice - Strategies of Speaking - Speaking about Scientists / Celebrities, Narrating the place of visit - Movie / book review - Compering an event - Delivering welcome address / Proposing vote of thanks . Activity: Conducting mock event.	
UNIT-V	FIELD STUDY	3+3
	Concept: Over view of field study - Objective(s) of the survey - Methodology - Designing a questionnaire - field survey / interview techniques - Collection of data - Summarizing the data - Presentation . Activity: Based on certain specific objective(s), 3-5 persons in the society need to be interviewed - team event: 1/2/3 students per team; each team has to make a presentation.	
	Total (LT)	32 Periods
	OPEN ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the class room teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course the students will be able to:	BLOOM'S Taxonomy
CO1	Demonstrate an understanding of grammatical structures in conversations	L3 – Apply
CO2	Apply the strategies of skimming and scanning to comprehend the text.	L3 – Apply
CO3	Develop writing skills in a professional context.	L3 - Apply
CO4	Use correct intonation to enhance public speaking skills.	L3 - Apply
CO5	Build interpersonal skills to perform well in an interview.	L3 - Apply
	TEXTBOOKS:	
1.	English for Engineers & Technologists Orient Blackswan Private Ltd. Department of English, Anna University, Chennai. 1999.	
	REFERENCE BOOKS:	
1.	Raman. Meenakshi, & Sangeeta Sharma. Professional English. Oxford UP : New Delhi, 2019.	
2.	Arora V.N. and Laxmi Chandra. Improve Your Writing. Oxford Univ. Press : New Delhi, 2001.	
3.	Chellammal. V. Learning to Communicate. Allied Publishers : New Delhi, 2003.	
4.	Kumar, Kulbhusan and RS Salaria. Effective Communication Skill. Khanna Publishing House : New Delhi, 2016.	

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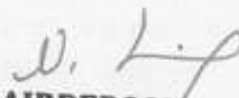
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5.	Lewis, Norman. Word Power Made Easy. Goyal Publishers Pvt., Ltd.: New Delhi, 2020			
S.No.	ONLINE COURSES:			
1.	https://www.totalsuccess.co.uk/online-letter-writing-course/			
2.	https://onlinecourses.nptel.ac.in/noc23_hs115/preview			
	WEB REFERENCES:			
S.No.	Publisher	Website link		Type of Content
1.	Grammarly	https://the FluentLife.com/content/steps-to-learn-english-grammar-easily/		Grammar
2.	Grammarly	https://www.grammarly.com/grammar#sectionGroup_6iKEWxDNd9GIgyj522RuVP		Grammar
	VIDEO REFERENCES:			
S.No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	Tenses	Perfect English Grammar	Grammar	https://www.perfect-englishgrammar.com/learn-english-video.html
2.	Conditional Clauses, Prepositions and Reported Speech	English Language for Competitive Exams	Grammar	https://www.youtube.com/watch?v=TMYTIL79BWw

COs	POs											PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1								1	3		1			
CO2								1	3		1			
CO3								1	3		1			
CO4								1	3		1			
CO5								1	3		1			
Average								1.0	3.0		1.0			
1-Low, 2 -Medium, 3-High.														


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BE23MA210	LINEAR ALGEBRA AND NUMERICAL METHODS	CP	L	T	P	C
		3	2	1	0	3
Programme & Branch	B.E. – Electronics and Computer Engineering	Version: 1.0				
Course Objectives:						
1.	To characterize the parameters of linear system of equations in vector space.					
2.	To familiarize the concepts of linear transformations between the vector spaces.					
3.	To enable students to understand and apply fundamental methods to solve equations.					
4.	To provide students with an understanding of Interpolation techniques.					
5.	To make the students to understand and apply single and multistep methods for solving first order ordinary differential equations.					
	INTRODUCTION (Not for Examination)					2
Importance	Importance: Linear algebra are foundational pillars in the field of Electronics and Communications Engineering, providing essential tools for analyzing and designing systems in diverse applications such as Circuit Analysis, signal processing, and computer Graphics & Robotics.					
Real-life Example(s)	Image Compression Using JPEG, SPICE Circuit Simulation, Machine Learning					
Linkages	-					
UNIT-I	VECTOR SPACES					6+3
	Vector spaces – Subspaces – Linear combinations and linear system of equations – Linear independence and linear dependence – Bases and dimensions .					
UNIT-II	LINEAR TRANSFORMATION AND DIAGONALIZATION					6+3
	Linear transformation - Null spaces and ranges - Dimension theorem - Matrix representation of a linear transformation – Eigenvalues and eigenvectors – Diagonalizability .					
UNIT-III	SOLUTION OF EQUATION AND EIGENVALUE PROBLEMS					6+3
	Essential of Solution of Equations - Fixed point iteration method – Newton Raphson method – Solution of linear system of equations – Gauss elimination and Jordan methods – Iterative methods of Gauss Jacobi and Gauss Seidel - Eigenvalues of a matrix by Power method .					

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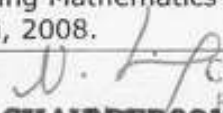
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UNIT – IV	APPROXIMATE SOLUTION TECHNIQUES	6+3
	A view on Interpolation - Lagrange's and Newton's forward and backward difference interpolations - Derivative of Newton's forward and backward difference interpolation - Problems - Numerical single and double integration using Trapezoidal and Simpson's 1/3 rules - Problems .	
UNIT-V	NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS	6+3
	Single step methods: Taylor's series method - Problems - Euler's method - Modified Euler's method - Fourth order Runge - Kutta method for solving first order differential equations - Problems - Multi step methods: Milne's predictor corrector methods for solving first order differential equations - Problems.	
	Total (LT)	47 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Apply the concepts of Vector Space to solve linear system of equations.	L3 – Apply
CO2	Solve and relate Eigen value and Eigen vectors using Linear Transformation and Diagonalization.	L3 – Apply
CO3	Apply direct and iterative methods for solving equations.	L3 – Apply
CO4	Identify and apply interpolation technique on Engineering applications.	L3 – Apply
CO5	Solve the solution of initial value problems using single and multi-step methods.	L3 – Apply
	TEXTBOOKS:	
1.	Friedberg. A.H., Insel. A.J. and Spence. L., "Linear Algebra", Prentice Hall of India, New Delhi, 4th Edition, 2004.	
2.	Grewal, B.S., and Grewal, J.S., "Numerical Methods in Engineering and Science", 10 th Edition, Khanna Publishers, New Delhi, 2015.	
	REFERENCE BOOKS:	
1.	Kreyszig E., "Advanced Engineering Mathematics", Tenth Edition, John Wiley and sons, 2011.	
2.	Ramana B.V., "Higher Engineering Mathematics", Sixth Edition, Tata McGraw Hill Publishing Company, New Delhi, 2008.	

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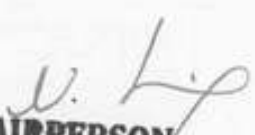
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WEB REFERENCES:				
S.No.	Publisher	Website link		Type of Content
1.	The International Linear Algebra Society	https://journals.uwyo.edu/index.php/ela/article/view/7783/6463		Journal
2.	Tutorials point	https://ncas.in/wp-content/uploads/2024/08/K.ALAMELU-PPT-OF-VECTOR-SPACE.pdf		Lecture Notes
VIDEO REFERENCES:				
S.No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	NPTEL	Prof. Dilip Patil	LECTURE	https://onlinecourses.nptel.ac.in/noc22_ma45/preview
2.	NPTEL	Prof. Ameeya Kumar Nayak	LECTURE	https://onlinecourses.nptel.ac.in/noc19_ma21/preview

Mapping of COs with POs and PSOs														
COs	POs											PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2												
CO2	3	2												
CO3	3	2												
CO4	3	2												
CO5	3	2												
Avg.	3.0	2.0												
1-Low, 2 -Medium, 3-High.														


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BE23GE303	ENGINEERING GRAPHICS AND CIRCUIT DRAWINGS	CP	L	T	P	C
		5	1	0	4	3
Programme & Branch	COMMON TO B.E. (ECE, EEE AND ECX) BRANCHES	Version: 1.1				
Use of A3 sheets and Drawing Instruments are Permitted						
Course Objectives:						
1.	To understand the importance of basic concepts and principles of engineering drawing					
2.	To develop the ability to communicate with others through technical drawings and sketching					
3.	To create simple engineering designs of industrial components using CAD software					
4.	To enable the knowledge about the components and its forms of interpretation of graphics					
5.	To understand the basics of Electrical and Electronics symbols and drawings					
	INTRODUCTION (Not for Examination)					2
Importance	Engineering graphics is a visual language that enables engineers to communicate complex ideas and designs precisely and unambiguously. It helps students visualize 3D objects from 2D views, improving spatial ability and design thinking. For electronics students, it directly supports schematic, PCB layout, and enclosure design documentation used in electronic product development.					
Real-life Example(s)	The LG 65LF6350-UA LED Smart TV service manual includes block diagrams and full circuit schematics for the power supply (SMPS), main board, and backlight driver, using standardized electronic symbols and signal annotations. It also provides PCB layouts and connector pin-out drawings that show physical board outline, mounting holes, and connector positions, which can be directly linked to orthographic/engineering views of the TV chassis.					
UNIT-I	GEOMETRIC CONSTRUCTION					3+12
	Introduction to Engineering Drawing, Lettering, Dimensioning, Drawing instruments, Sheet Layout, Drawing Standards (BIS) - Basic Geometrical constructions, Conic Sections - Construction of Ellipse, Parabola and Hyperbola by using eccentric method, Special Curves - Construction of Cycloid, Construction of Epicycloid, Construction of Hypocycloid					
UNIT-II	PROJECTION OF POINTS, LINES AND PLANE SURFACES					3+12
	Points using first angle projection and third angle projection, Projection of Straight Lines inclined to both the planes (only first angle projection) by using rotating line method- Projection of Planes (polygonal and circular surfaces) inclined to both principal planes by rotating object method					
UNIT-III	PROJECTION OF SOLIDS AND SECTION OF SOLIDS					3+12
	Projection of simple solids like Prism, Pyramid, Cylinder and Cone when the axis is inclined to one principal plane and parallel to other by rotating object method - Sectioning of solids (Prism, Pyramid, Cylinder and Cone) in simple vertical position when the cutting plane is inclined to one principal plane and perpendicular to the other and obtaining the true shape of the section					

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UNIT – IV	DEVELOPMENT OF SURFACES AND ISOMETRIC PROJECTIONS	3+12
	Development of lateral surfaces of simple sectioned solids (Prism, Pyramid, Cylinder and Cone) - Principles of Isometric Projection- Construction of Isometric Views of Prism, Pyramid, Cylinders and Cones- Combination of two solid objects in a simple vertical position	
UNIT – V (a)	FREE HAND SKETCHING AND ELECTRICAL AND ELECTRONICS CIRCUITS	2+9
	Visualization Concepts and Free hand sketching - Free hand sketching of multiple views from pictorial views of object - Exercise on Electrical Wiring Drawings and Electronics Circuit Drawings	
UNIT – V (b)	APPLICATIONS (Not for Examination)	4
	Study of Industrial Electrical Drawings - Study of Electrical Circuit Drawings - Study of Commercial Software Packages related EEE and ECE	
	Total(L+P)	75 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as IA only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Use BIS Standards in Engineering Drawing	L2 - Understand
CO2	Construct two dimensional drawing for Engineering applications	L3 - Apply
CO3	Construct projection of points, lines and planes	L3 - Apply
CO4	Visualize geometric solids and isometric projections	L3 - Apply
CO5	Construct the Electrical and Electronic Symbols and Circuits	L2 - Understand
	TEXTBOOKS:	
1.	Venugopal K and Prabhu Raja V, Engineering Graphics, New AGE International Publishers, 2018	
2.	Natarajan.K.V, A Textbook of Engineering Graphics, Dhanalakshmi Publishers, Chennai, 2015	
	REFERENCE BOOKS:	
1.	Basant Agrawal, Agrawal C.M., "Engineering Drawing", Second Edition, McGraw Hill Education, 2019	
2.	Gopalakrishnana K.R. "Engineering Drawing", Volume. I & II, Subhas Publications, Bengaluru, 2014	
3.	Parthasarathy N.S., Vela Murali. "Engineering Drawing", First Edition, Oxford University Press, 2015	

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WEB REFERENCES:				
S. No.	Publisher	Website link		Type of Content
1.	NPTEL	https://nptel.ac.in/courses/112103019		Tutorials and material
2.	Engineering Drawing Blog	www.engineeringdrawing.org/2012/04/solids-section		Articles
VIDEO REFERENCES:				
S. No.	Video Details	Name of the Expert	Type of Content	Video link
1.	Engineering Graphics and Design	Prof. Naresh Varma Datla	NPTEL	https://onlinecourses.nptel.ac.in/noc21_me128/preview

Mapping of COs with POs and PSOs											
COs	POs										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	1	2	2	2				3		2
CO2	3	1	2	2	2				3		2
CO3	3	1	2	2	2				3		2
CO4	3	1	2	2	2				3		2
CO5	3	1	2	2	2				3		2
Average	3.0	1.0	2.0		2.0				3.0		2.0
1-Low, 2 -Medium, 3-High.											

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BE23EC401	ELECTRONIC DEVICES	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	COMMON TO B.E. (ECE AND ECX) BRANCHES	Version: 1.1				
Course Objectives:						
1.	To describe the theory, operations, characteristics, and applications of semiconductor diodes					
2.	To understand theory, operation, and characteristics of the Bipolar junction transistor					
3.	To understand theory, operation, and characteristics of the Field effect transistors					
4.	To classify the construction, theory and, operation of the special semiconductor devices					
5.	To classify the construction and working principles of various power devices and display devices					
	INTRODUCTION (Not for Examination)					2
Importance	This course establishes foundational knowledge about the building blocks of modern technology, such as diodes and transistors. It provides an understanding on the basic operations that drives all computing hardware, from simple circuits to complex microprocessors.					
Real-life Example(s)	Smartphone Charger: The smartphone charger uses a rectifier circuit built with diodes to ensure AC power is converted to DC power needed to charge your battery.					
Linkages	Pre-Requisite : Nil Future courses : Analog Electronic Circuits, Digital Electronics and Microprocessor					
UNIT-I	SEMICONDUCTOR DIODES					9
	PN junction diode, forward and reverse bias characteristics, Transition and Diffusion Capacitances, Zener diode, Forward and reverse bias characteristics, Breakdown in PN Junction Diodes, Applications - Rectifier and voltage regulator.					
UNIT-II	BIPOLAR JUNCTION TRANSISTOR					9
	NPN -PNP -Operations-Early effect-Current equations — Input and Output characteristics of CE, CB, CC, h-parameter model, Hybrid and pi model – Eber's, Multi Emitter Transistor.					
UNIT-III	FIELD EFFECT TRANSISTORS					9
	JFETs — Drain and Transfer characteristics, -Current Equations-Pinch off voltage and its significance- MOSFET- Characteristics- D-MOSFET, E-MOSFET - Characteristics).					
UNIT - IV	SPECIAL SEMICONDUCTOR DEVICES					9
	Schottky barrier diode -Varactor diode -Tunnel diode - LASER Diode, LDR, Photo Diode, Photo transistor, Solar cell.					
UNIT - V	POWER DEVICES AND DISPLAY DEVICES					9
	SCR,DIAC, TRIAC, IGBT, Light Emitting Diode (LED) and its types , Liquid Crystal Diode (LCD) and its types.					
	Total(L)					47 Periods

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OPEN-ENDED PROBLEMS / QUESTIONS				
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as IA only and not for the End semester Examinations.			
	Course Outcomes: Upon completion of this course, the students will be able to:			BLOOM'S Taxonomy
CO1	Understand the basics of semiconductor diodes, operations, characteristics, and applications			L2-Understand
CO2	Understand the operation, characteristics, and modeling of BJT			L2-Understand
CO3	Understand the operation, characteristics, and modeling of FET			L2-Understand
CO4	Classify the construction and working principles of special semiconductor devices			L2-Understand
CO5	Classify the construction and working principles of power devices and display devices			L2-Understand
	TEXTBOOKS:			
1.	Donald A Neaman, "Semiconductor Physics and Devices", 4 th Ed., McGrawHill, 2021 (Special Indian Edition)			
2.	S.M. Sze, M.K. Lee, "Semiconductor Physics And Devices (An Indian Adaptation)", 3 rd Ed., Wiley 2021			
	REFERENCE BOOKS:			
1.	Robert Boylestad and Louis Nashelsky, "Electron Devices and Circuit Theory", 10 th Ed., Pearson Prentice Hall, July 2008			
2.	R.S.Sedha, "A Text Book of Applied Electronics" S.Chand Publications, 2006			
3.	Salivahanan. S, Suresh Kumar. N, Vallavaraj.A, "Electronic Devices and circuits", 3 rd Ed., Tata McGraw Hill, 2008.			
	WEB REFERENCES:			
S. No.	Publisher	Website link		Type of Content
1.	Electronics Tutorials	electronics-tutorials.com		Tutorials and material
2.	All About Circuits	allaboutcircuits.com		Articles
	VIDEO REFERENCES:			
S. No.	Video Details	Name of the Expert	Type of Content	Video link
1.	Basic Electronics	Prof. T.S. Natarajan	NPTEL	https://l1nq.com/gXBSe
2.	Physics of Semiconductor Devices	Prof. C.R. Selvakumar	NPTEL	nptel.ac.in/courses/115106059/

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	Mapping of COs with POs and PSOs										
COs	POs										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	1									
CO2	2	1									
CO3	2	1									
CO4	2	1									
CO5	2	1									
Average	2.0	1.0									
	1-Low, 2 -Medium, 3-High.										



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BE23MC902	தமிழரும் தொழில்நுட்பமும் / Tamils and Technology	CP	L	T	P	C
		1	1	0	0	1
Programme & Branch	Common to all B.E. / B.Tech. Branches	Version: 1.1				
Students can write the examination either in Tamil or in English						
Course Objectives:						
1.	To acquire knowledge of technology during the Sangam age.					
2.	To learn about house furnishings, sculptures and temple design during the Sangam age.					
3.	To develop knowledge of metallurgical studies as a source of historical and archaeological evidence.					
4.	To acquire knowledge of ancient techniques used in agriculture and agro-processing.					
5.	To know the development of Tamil through computer and to develop knowledge of Tamil.					
	INTRODUCTION (Not for Examination)					1
Importance	Includes artifacts, sculptures, temple designs during the Sangam period, historical and archaeological evidence, metallurgical studies, agriculture, ancient industrial techniques, and the development of Tamil through computer science.					
UNIT-I	WEAVING AND CERAMIC TECHNOLOGY					3
	Weaving Industry during Sangam Age - Ceramic technology - Black and Red Ware Potteries (BRW) - Graffiti on Potteries.					
UNIT-II	DESIGN AND CONSTRUCTION TECHNOLOGY					3
	Designing and Structural construction House & Designs in household materials during Sangam Age - Building materials and Hero stones of Sangam age - Details of Stage Constructions in Silappathikaram - Sculptures and Temples of Mamallapuram - Great Temples of Cholas and other worship places - Temples of Nayaka Period - Type study (Madurai Meenakshi Temple)- Thirumalai Nayakar Mahal - Chetti Nadu Houses, Indo - Saracenic architecture at Madras during British Period.					
UNIT-III	MANUFACTURING TECHNOLOGY					3
	Art of Ship Building - Metallurgical studies - Iron industry - Iron smelting, steel - Copper and gold Coins as source of history - Minting of Coins - Beads making - industries Stone beads - Glass beads - Terracotta beads - Shell beads / bone beats - Archeological evidences - Gem stone types described in Silappathikaram .					

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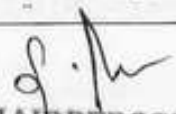
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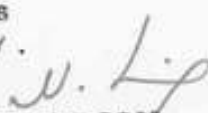
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UNIT – IV	AGRICULTURE AND IRRIGATION TECHNOLOGY	3
	Dam, Tank, ponds, Sluice, Significance of Kumizhi Thoompu of Chola Period, Animal Husbandry - Wells designed for cattle use - Agriculture and Agro Processing - Knowledge of Sea - Fisheries - Pearl - Conche diving - Ancient Knowledge of Ocean - Knowledge Specific Society .	
UNIT-V	SCIENTIFIC TAMIL & TAMIL COMPUTING	3
	Development of Scientific Tamil - Tamil computing - Digitalization of Tamil Books - Development of Tamil Software - Tamil Virtual Academy - Tamil Digital Library - Online Tamil Dictionaries - Sorkuvai Project .	
	Total	16 Periods
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Recognize the importance of technology in supporting food production, shelter, clothing, and transport during the Sangam Age	L1 - Remember
CO2	Classify the types of furnishings, sculptures, and temple designs used during the Sangam age.	L2 - Understand
CO3	Summarize how metallurgical evidence contributes to historical and archaeological studies.	L2 - Understand
CO4	Recognize the role of these techniques in ensuring food security and sustainable agriculture	L1 - Remember
CO5	Illustrate the role of digital tools in developing Tamil.	L2 - Understand
	TEXTBOOKS:	
1.	டாக்டர் கே.கே.பிள்ளை, "தமிழக வரலாறு மக்களும் பண்பாடும்", (வெளியீடு, தமிழ்நாடு பாடநூல் கல்வியியல் பணிகள் கழகம்).	
2.	முனைவர் இல.சுந்தரம், "கணினித்தமிழ்", (விகடன் பிரசுரம்).	
	REFERENCE BOOKS:	
1.	"கீழடி - வைகை நதிக்கரையில் சங்ககால நகர நாகரிகம்", (தொல்லியல் துறை வெளியீடு).	
2.	"பொருதை - ஆற்றங்கரை நாகரிகம்", (தொல்லியல் துறை வெளியீடு).	

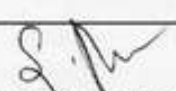

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WEB REFERENCES:				
S.No.	Publisher	Website link		Type of Content
1.	World Architecture Community	https://worldarchitecture.org/architecture-news/eepe/ancient-construction-techniques-of-india-a-regional-study.html		e-resources
2.	Poornaprajna Publication, India.	383667806_Ancient_Indian_Agriculture Technology" https://www.researchgate.net/publication/383667806_Ancient_Indian_AgricultureTechnology		e-resources
VIDEO REFERENCES:				
S.No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	YouTube	Dr. Anila Varghese	Lectures	https://youtu.be/JmvaLaAnSrI?si=kQkYfk3GS82Mqm59
2.	YouTube	Suryan FM	Lectures	https://youtu.be/W5LcPzfkFMo?si=B4bEpIgg3KolhExx

Mapping of COs with POs and PSOs														
COs	POs											PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	1										1			
CO2							1				2			
CO3						2	1				2			
CO4					2	2	1							
CO5					2						2			
Avg.	1.0				2.0	2.0	1.0				1.75			
1-Low, 2 -Medium, 3-High.														


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BE23MC903	UNIVERSAL HUMAN VALUES AND ETHICS	CP	L	T	P	C
		3	2	1	0	3
Programme & Branch	COMMON TO ALL B.E. / B.TECH. BRANCHES	Version: 1.1				
Course Objectives:						
1.	To understand the concept of Universal Human Values.					
2.	To discuss theoretical and practical implications of UHV.					
3.	To relate the use of harmony in the family and society.					
4.	To classify the harmony in the nature methods.					
5.	To construct effective human values in personal and professional in life.					
	INTRODUCTION (Not for Examination)					2
Importance	Importance: Universal human values and ethics are foundational to build a harmonious and peaceful society. These values-such as respect, kindness, empathy, fairness, honesty, and responsibility-serve as a moral compass for individuals and communities.					
Real-life Example(s)	Analysis of various societal issues, environmental issues and find solution.					
Linkages	-					
UNIT-I	INTRODUCTION TO VALUE EDUCATION					6+3
	Right Understanding , Relationship and Physical Facility (Holistic Development and the Role of Education) - Understanding Value Education - Sharing about Oneself - Self-exploration as the Process for Value Education - Continuous Happiness and Prosperity - the Basic Human Aspirations - Exploring Human Consciousness - Happiness and Prosperity - Current Scenario - Method to Fulfil the Basic Human Aspirations - Exploring Natural Acceptance .					
UNIT-II	HARMONY IN THE HUMAN BEING					6+3
	Understanding Human being as the Co-existence of the Self and the Body - Distinguishing between the Needs of the Self and the Body - Exploring the difference of Needs of Self and Body - The Body as an Instrument of the Self - Understanding Harmony in the Self - Exploring Sources of Imagination in the Self - Harmony of the Self with the Body - Programme to ensure self-regulation and Health - Exploring Harmony of Self with the Body .					
UNIT-III	HARMONY IN THE FAMILY AND SOCIETY					6+3
	Harmony in the Family - the Basic Unit of Human Interaction - 'Trust' - the Foundational Value in Relationship - Exploring the Feeling of Trust - 'Respect' - as the Right Evaluation - Exploring the Feeling of Respect - Other Feelings , Justice in Human-to-Human Relationship					

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	Understanding Harmony in the Society - Vision for the Universal Human Order - Exploring Systems to fulfil Human Goal .	
UNIT – IV	HARMONY IN THE NATURE / EXISTENCE	6+3
	Understanding Harmony in the Nature – Interconnectedness , self-regulation and Mutual Fulfilment among the Four Orders of Nature - Exploring the Four Orders of Nature - Realizing Existence as Co-existence at All Levels - The Holistic Perception of Harmony in Existence - Exploring Co-existence in Existence .	
UNIT – V	IMPLICATIONS OF THE HOLISTIC UNDERSTANDING - A LOOK AT PROFESSIONAL ETHICS	6+3
	Natural Acceptance of Human Values - Definitiveness of (Ethical) Human Conduct - Exploring Ethical Human Conduct - A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order - Competence in Professional Ethics - Exploring Humanistic Models in Education - Holistic Technologies, Production Systems and Management Models -Typical Case Studies - Strategies for Transition towards Value-based Life and Profession - Exploring Steps of Transition towards Universal Human Order .	
	Total	47 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Interpret the concepts of Universal Human Values.	L2 - Understand
CO2	Summarize both theoretical and practical implications of Universal Human Values.	L2 - Understand
CO3	Build the harmony in family and society.	L3 - Apply
CO4	Practice harmony in all human existence.	L3 - Apply
CO5	Relate human values in both personal and professional life.	L2 - Understand
	TEXTBOOKS:	
1.	R R Gaur, R Asthana, G P Bagaria, A Foundation Course in Human Values and Professional Ethics, 2nd Revised Edition, Excel Books, New Delhi, 2019.	
2.	A.N. Tripathi, Human Values, New Age Intl. Publishers, New Delhi, 2004.	
	REFERENCE BOOKS:	
1.	R.R Gaur, R Sangal, G P Bagaria, A foundation course in Human Values and professional Ethics – Teachers Manual, Excel books, New Delhi, 2010.	

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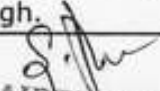
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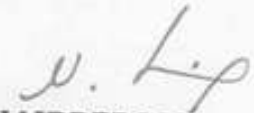
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2.	B L Bajpai, 2004, Indian Ethos and Modern Management, New Royal Book Co., Lucknow, Reprinted 2008.		
3.	Frankl, Viktor E. Yes to Life In spite of Everything, Penguin Random House, London, 2019.		
4.	Van Zomeren, M., & Dovidio, J. F. The Oxford Handbook of the Human Essence (Eds.), New York Oxford University Press, 2018.		
WEB REFERENCES:			
1.	https://www.researchgate.net/publication/382495858_Universal_Human_Value		
2.	https://fdp-si.aicte-india.org/UHVII.php		
VIDEO REFERENCES:			
S. No.	Publisher	Website link	Type of Content
1.	Dr.Rajneesh Arora	https://www.youtube.com/watch?v=SdXIB0k7klQ&list=PLWDeKF97v9SP7wSlapZcQRrT7OH0ZIGC4&index=2	Video Lecture
2.	Mr.Umesh Jadhav	https://www.youtube.com/watch?v=_g8Hw2APx8Q	Video Lecture

Mapping of COs with POs and PSOs

COs	POs											PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1						2					2			
CO2							2							
CO3						3								
CO4							3				2			
CO5						3		2						
Avg.						2.6	2.5	2.0			2.0			
1-Low, 2 -Medium, 3-High.														


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 Approved in Academic Council Meeting held on 11/10/2025

BE23GE310	OBJECT ORIENTED PROGRAMMING USING C++	CP	L	T	P	C
		5	3	0	2	4
Programme & Branch	Common to B.E. (CSE, ECX) and B.TECH. (IT, CSBS and AI&DS) Branches	Version: 1.0				
Course Objectives:						
1.	To understand the principles of object-oriented programming concepts.					
2.	To apply the concept classes, objects and encapsulation.					
3.	To explore the inheritance and abstract classes.					
4.	To illustrate the polymorphism.					
5.	To develop the applications with exception handlers.					
6.	To apply various, I/O techniques for console and file I/O.					
	INTRODUCTION (Not for Examination)					2
Importance	In today's digital world, software runs almost everything such as mobile phones, banking systems, cars, satellites, and medical devices. To create such reliable and efficient software, powerful programming languages are needed, and C++ is one of the most widely used and important languages.					
Real-life Example(s)	E-Commerce Applications, Banking System, College Management System, Employee Management System.					
Linkages	Pre-requisite: Problem Solving using C Programming. Future courses: Java Programming, C# and Dot Net.					
UNIT-I	BASICS OF C++ PROGRAMMING					9
	Limitation of Structure Oriented Programming - Need of Object-Oriented Programming - C++ Introduction - Structure of C++ Programming - Compiling - Executing and Debugging - Character Set(L2) - Tokens: (Keywords - Identifiers - Constants - Strings - Operators - Special Symbols) - Data Types. Expression(L2) - Precedence and Associativity - Evaluating Expression - Type Conversion - Input and Output - Control Flow Statements - Arrays - Functions - Inline Functions - Default Arguments.					
UNIT-II	CLASS, OBJECTS AND ENCAPSULATION					9
	Class Definition - Access Specifiers - Object Creation - Array of Objects - Constructor - Destructor - this Pointer - Static variables and Member Functions - Encapsulation: Introduction - Types (21) - Friend function and Friend class.					
UNIT-III	INHERITANCE AND ABSTRACT CLASS					9
	Inheritance: Needs - types of inheritance - Constructors and					

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	Destructors in Inheritance - Constraints of Multiple Inheritance - Abstract Base Class - Pure Virtual function.	
UNIT-IV	POLYMORPHISM	9
	Polymorphism: Introduction - Compile Time polymorphism: Function Overloading - Operator Overloading - Run Time Polymorphism - Function Overriding - Virtual Function.	
UNIT-V	EXCEPTION HANDLING AND IO STREAMS	9
	Exception Handling: Needs - try - catch - throw - Handling any type of Exceptions - User type of Exceptions. I/O Streams - Manipulators - Overloading Inserters (<<) and Extractors (>>) - Sequential and Random files - binary files.	
	Total (L)	47
		Periods
	LIST OF EXERCISES:	
1.	Write a C++ program to sort an array of elements using functions.	
2.	Write a C++ program to demonstrate call by value and call by reference.	
3.	Write a C++ program to specify default arguments.	
4.	Write a program Illustrating class declarations, definition, and accessing class members.	
5.	Write a program to illustrate default constructor, parameterized constructor and copy constructors.	
6.	Write a program to demonstrate Friend Function and Friend Class.	
7.	Write a program to demonstrate binary Operator Overloading.	
8.	Write C++ programs that illustrate how the following forms of inheritance are supported: a) Multiple inheritance b) Multi level inheritance.	
9.	Write a Template based program to Sort the Given List of Elements.	
10.	Write a Program to demonstrate the Catching of All Exceptions.	
11.	Write a program to illustrate Abstract Class.	
12.	Write a C++ program to demonstrate virtual function.	
	Total (P)	30
		Periods
	Total (L+P)	77
		Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open-Ended Problems will be solved during the	

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	classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.			
	Course Outcomes: Upon completion of this course, the students will be able to:			BLOOM'S Taxonomy
CO1	Apply the concepts of object - oriented programming.			L2 - Understand
CO2	Examine the use of objects and encapsulation to solve the real-world problems.			L3 - Apply
CO3	Utilize the code reusability for critical applications.			L3 - Apply
CO4	Implement the real-time applications with polymorphism.			L3 - Apply
CO5	Demonstrate the use of exception handling.			L3 - Apply
TEXTBOOKS:				
1.	Balagurusamy, "Object Oriented Programming with C++", 8 th Edition, McGraw Hill, 2020.			
2.	Venugopal.K.R., Raj Buyya, "Mastering C++", 2 nd Edition, Tata Mcgraw Hill, 2017.			
3.	Bjarne Stroustrup, "The C++ Programming Language", 4 th Edition, Addison-Wesley, 2013.			
REFERENCE BOOKS:				
1.	Herbert Schildt, "C++: The Complete Reference", 5 th Edition, McGraw Hill Education, 2012.			
2.	Balagurusamy E, "Object Oriented Programming with C++", 8 th Edition, Tata McGraw-Hill, NewDelhi, 2019.			
WEB REFERENCES:				
S.No	Publisher	Website link		Type of Content
1.	cplusplus.com	https://cplusplus.com/forum/beginner/165465/		Web Content
2.	geeksforgeeks	https://www.geeksforgeeks.org/object-oriented-programming-in-cpp/		Web Content
3.	Learn C++	https://www.learncpp.com/cpp-tutorial/welcome-to-object-oriented-programming/		Web Content
VIDEO REFERENCES:				
S.No	Video Details	Name of the Expert	Type of Content	Video Link
1.	YouTube	freeCodeCamp.org	Demonstration	https://www.youtube.com/watch?v=vLnPwxZdW4Y
2.	YouTube	freeCodeCamp.org	Demonstration	https://www.youtube.com/watch?v=wN0x9eZLix4

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3.	YouTube	thenewboston	Demonstration	https://www.youtube.com/watch?v=tvC1WCdV1XU
4.	YouTube	Sloan Kelly	Demonstration	https://www.youtube.com/watch?v=0Zr_0Jy8mWE
ONLINE COURSES:				
1.	Udemy - "Learn Advanced C++ Programming"			
2.	Coursera - "Object-Oriented Data Structures in C++"			
3.	pluralsight - "C++ Fundamentals Including C++ 17"			
4.	edX - "Object-Oriented Programming in C++"			
5.	Codecademy - "Learn C++"			

Mapping of COs with POs and PSOs														
COs	POs											PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	2	2					1		2			
CO2	3	2	2	2					1		2			
CO3	3	2	2	2					1		2			
CO4	3	2	2	2					1		2			
CO5	3	2	2	2					1		2			
Avg.	3.0	2.0	2.0	2.0					1.0		2.0			
1-Low, 2 -Medium, 3-High.														

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BE23EC402	CIRCUIT THEORY AND ANALYSIS	CP	L	T	P	C
		5	3	0	2	4
Programme & Branch	COMMON TO B.E. (ECE AND ECX) BRANCHES	Version: 1.1				
Course Objectives:						
1.	To apply basic electric laws and methods in DC and AC circuits analysis					
2.	To apply network theorems for solving the electric circuits					
3.	To examine the transient and steady-state response of the circuits by applying DC and AC excitations					
4.	To construct and determine the responses of combinations of R, L and C circuits					
5.	To construct two – port networks and for finding the various parameters					
	INTRODUCTION (Not for Examination)					2
Importance	This course is the bedrock for an electronics student. It is crucial for designing anything from a simple power supply to complex integrated circuits. Circuit theory is indispensable for design and implementation of innovative electronic systems.					
Real-life Example(s)	Electric ceiling fan: Adjusting speed of the fan uses Impedance control. Changing regulator speed 5 to speed 1, adds more impedance to the circuit. This added impedance limits the current flow to the fan motor, while changing the voltage delivered across the motor terminals that lead to lower fan speed.					
Linkages	Pre-Requisite : Nil Future courses : Analog Electronic Circuits, Digital Electronics and Microprocessor					
UNIT-I	CIRCUIT ANALYSIS					9
	Basic Components of electric Circuits- Voltage and Current Sources- Ohms Law - Kirchhoff's Current Law - Kirchhoff's voltage law - Series and Parallel Connected Independent Sources - Resistors in Series and Parallel - Voltage and current division - Mesh current and node voltage method of analysis for DC circuits.					
UNIT-II	NETWORK THEOREMS					9
	Wye- delta conversion – Thevenin's and Norton's theorems – Superposition theorem – Maximum power transfer theorem – Reciprocity theorem.					
UNIT-III	SINUSOIDAL STEADY STATE ANALYSIS					9
	Sinusoidal Steady – State analysis, Phasor relationship for R, L and C - Impedance and Admittance - Mesh and Nodal analysis for AC circuits - AC Circuit Power Analysis - Instantaneous Power - Average Power - Apparent Power and Power Factor - Complex Power.					

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UNIT – IV	TRANSIENTS AND RESONANCE IN RLC CIRCUITS	9
	Basic RL and RC Circuits - The Source- Free RL Circuit - The Source-Free RC Circuit - The Unit-Step Function - Driven RL Circuits - Driven RC Circuits - RLC Circuits - Frequency Response - Parallel Resonance - Series Resonance - Quality Factor.	
UNIT – V	COUPLED CIRCUITS AND TWO-PORT NETWORKS	9
	Magnetically Coupled Circuits - Self and Mutual Inductance - Linear Transformer - Ideal Transformer - Two port network Parameters: Impedance - Admittance - Transmission and Hybrid - Relation between parameters.	
	Total(L)	47 Periods
	LIST OF EXPERIMENTS/EXERCISES:	
1.	Verifications of KVL and KCL.	
2.	Verification of Thevenin and Norton Theorems.	
3.	Verification of Superposition Theorem.	
4.	Verification of Maximum power transfer Theorem.	
5.	Determination of Resonance Frequency of Series and Parallel RLC Circuits.	
6.	Transient analysis of RL and RC circuits.	
	Total(P)	30 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as IA only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Apply the basic concepts of circuit analysis such as Kirchoff's laws, mesh current and node voltage method for analysis of DC and AC circuits	L3 – APPLY
CO2	Apply suitable network theorems to verify AC and DC circuits	L3 – APPLY
CO3	Apply various steady state analysis for R, L and C circuits	L3 – APPLY
CO4	Solve the transient and frequency response for RC, RL and RLC circuits	L3 – APPLY
CO5	Apply the concepts of coupled circuits and two-port networks to simple electronic circuits	L3 – APPLY
	TEXTBOOKS:	
1.	W. H. Hayt Jr., Jack E. Kemmerly, and S. M. Durbin, "Engineering Circuit Analysis", 9 th Ed., McGraw Hill, 2020	
2.	Abhijit Chakrabarti, "Circuit Theory (Analysis and Synthesis)", 7 th Ed., Dhanpat Rai Publications, 2018	
	REFERENCE BOOKS:	
1.	A. Sudhakar and Shyammoohan S. Palli, "Circuits and Networks: Analysis and Synthesis", 6 th Ed., McGraw Hill, 2025	

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2.	Charles K. Alexander and Matthew N. O. Sadiku, "Fundamentals of Electric Circuits", 7 th Ed., McGraw Hill, 2022			
3.	J. David Irwin, R. Mark Nelms, "Engineering Circuit Analysis", 11 th Ed., Wiley, 2018			
WEB REFERENCES:				
S. No.	Publisher	Website link		Type of Content
1.	All About Circuits	www.allaboutcircuits.com		Textbooks and articles
2.	ScienceDirect	www.sciencedirect.com		Book chapters, and references
VIDEO REFERENCES:				
S. No.	Video Details	Name of the Expert	Type of Content	Video link
1.	NPTEL	Prof. S. C. Dutta Roy (IIT Delhi)	Full Course from NPTEL	https://www.youtube.com/playlist?list=PL1D46B10238154408
2.	MIT – OCW	Prof. Anant Agarwal	Foundational course in circuits and electronics	https://www.youtube.com/watch?v=RsJ1eg7XNVs

Mapping of COs with POs and PSOs											
COs	POs										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2			2			3	2		
CO2	3	2			2			3	2		
CO3	3	2			2			3	2		
CO4	3	2			2			3	2		
CO5	3	2									
Average	3.0	2.0			2.0			3.0	2.0		
1-Low, 2 -Medium, 3-High.											

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BE23PT802	HUMAN EXCELLENCE AND VALUE EDUCATION - II	CP	L	T	P	C
		2	1	0	1	NC
Programme & Branch	COMMON TO ALL B.E. / B.TECH. BRANCHES	Version: 1.1				
Course Objectives:						
1.	To understand habit development and avoid bad habits for a happy and successful life.					
2.	To inculcate essential values and ethics.					
3.	To understand interpersonal skills for good communication.					
4.	To learn methods, tools, and techniques for effective presentations.					
5.	To know methods for effective teamwork.					
	INTRODUCTION (Not for Examination)					2
Importance	Holistic personality development is crucial for engineers, as technical skills alone are insufficient in today's dynamic and competitive environment. Engineers must become well-rounded professionals capable of applying their knowledge, expressing ideas clearly, collaborating with others, and demonstrating strong leadership. Ultimately, this development enables future engineers to lead meaningful lives and make valuable contributions to society.					
Real-life Example(s)	(i) Personal development helps individuals build confidence, improve skills, and make better decisions, leading to greater success and fulfillment. (ii) It is crucial for a team leader to use active listening and clear communication to resolve conflicts, helping her team work better and succeed. (iii) Presentation skills are important for engineers because they help clearly explain complex ideas, persuade stakeholders, and collaborate effectively, making their work more impactful. (iv) Teamwork brings diverse skills and perspectives together, fosters collaboration across cultures, and drives innovation to achieve common goals efficiently.					
Linkages	Human Excellence and Value Education - I & Human Excellence and Value Education - III					
UNIT-I	HABITS FOR PERSONAL DEVELOPMENT					3+3
	Health Management - Becoming an effective adult and handling adolescent issues - Habit vs Addiction - Awareness of Human Physiology - Stay Away Habits : Smoking, Alcohol, Drugs, Violence - How to Handle Assaults : Physical, Emotional and Social - Cybercrimes - Awareness of Road Safety - Effective Habit Development : Yoga, Meditation, Sports and fitness, Sleep management, food and nutrition .					

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UNIT-II	VALUES AND ETHICS	3+3
	Values : Self-respect, Punctuality, Respecting Others Nonviolence, Truth, empathy, Honesty and integrity, Inner cleanliness - Defining Happiness - Encountering Failures, obstacles, Insults, Criticism - overcoming fear, jealousy hatred, Greed sorrow and anger - Desire management - Understanding Indian Culture & its Scientific Heritage .	
UNIT- III	INTERPERSONAL SKILLS	3+3
	Types of Relationships - Factors influencing Relationships - Barriers in Relationship Management - Best Practices for Relationship Management - Effective usage of EQ in Relationship Management - Understanding Personalities and Style Flexing .	
UNIT - IV	PRESENTATION SKILL	3+3
	Concepts: Occasions - Effect Voice Management - Elements of Presentation - Developing effective presentation - Delivering an effective presentation . Activities: Preparing and Delivering Presentation	
UNIT-V	TEAMWORK	3+3
	Concepts: Understanding the Roles of a Team Builder - Team Manager and Team Player - How to bring Synergy - Dynamics, Bonding and Alignment - Best Team Member Qualities - Characteristics of High-Performance Teams - Art of Persuasion - Art of Influencing - Building Trust . Activities: Demonstrating an Activity as a Team	
	Total	32 Periods
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Overcome the influence of bad habits and develop good habits.	L2 - Understand
CO2	Practice the values and ethics and lead a happy and healthy life.	L2 - Understand
CO3	Demonstrate interpersonal skills and work with others effectively	L2 - Understand
CO4	Deliver effective presentations for better communication.	L2 - Understand
CO5	Work as a team for the successful completion of the projects	L2 - Understand

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TEXTBOOKS:				
1.	Trainer and Faculty Lecture Notes / PPT			
REFERENCE BOOKS:				
1.	Stephen R. Covey, "The 7 Habits of Highly Effective People: Powerful Lessons in Personal Change", Free Press, 2004.			
2.	James Clear, "Atomic Habits", Random House Business books, 2018			
3.	Suresh Kumar E, Sreehari P, Savithri J, "Communication Skills and Soft Skills, Pearson India Education Services", 2011.			
4.	Alex K, "Soft Skills Know yourself and know the world", S. Chand & Company Pvt. Ltd., 2014.			
5.	Dale Carnegie, "The Art of Public Speaking", Rupa Publications India, 2018.			
6.	John C. Maxwell, "Teamwork 101: What Every Leader Needs to Know", HarperCollins Leadership, 2009.			
7.	Christopher Avery, "Teamwork Is an Individual Skill", Read How You Want, 2011.			
WEB REFERENCES:				
S.No.	Publisher	Website link	Type of Content	
1.	AICTE - SIP	https://www.youtube.com/watch?v=OgdNx0X923I&list=PLYwzG2fd7hzc4HerTNkc3pS_IvcCfKznV	Universal Human Values	
2.	Learn By Watch	https://www.youtube.com/watch?v=XkB8mclNeSI	Interpersonal Skills	
3.	mrSKYee	https://www.youtube.com/watch?v=boCf3iY8qj8	Teamwork	
VIDEO REFERENCES:				
S.No.	Video Details	Name of the Expert	Type of Content	Video link
1.	Morals vs Ethics	PNG Guy	Memes	https://www.youtube.com/watch?v=VqfWt20S4rI
2.	The Ultimate Guide to Expert Interpersonal Skills	Vanessa Van Edwards	Video content	https://www.youtube.com/watch?v=Ib09GqWP5rY&t=29s
3.	Good Teamwork	Gerrit Maassen van den Brink	Video content	https://www.youtube.com/watch?v=fUXdrl9ch_Q

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	ONLINE COURSES:	
1.	NPTEL Course on Enhancing Soft Skills and Personality - https://nptel.ac.in/courses/109104115	
2.	NPTEL course on Soft skills - https://nptel.ac.in/courses/109107121	

Mapping of COs with POs and PSOs														
COs	POs											PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1								2						
CO2										2	3			
CO3								2						
CO4							2	1	2					
CO5						2	2		2					
Avg.						2.0	2.0	1.7	2.0	2.0	3.0			
1-Low, 2 -Medium, 3-High.														



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B.E./B.Tech. Regulations 2023

BE23PT804		ENGINEERING CLINIC - I				CP	L	T	P	C
						2	0	0	2	1
(COMMON TO ALL BRANCHES)										
Programme & Branch		B.E. –Electronics and Computer Engineering					Version: 1.1			
Course Objectives:										
1	To understand the basics of real-world applications.									
2	To enable students to design, fabricate and demonstrate of a given application using PCB.									
3	To take entrepreneurship, product development, startup-related activities and problem-solving skills in higher semesters and final semester project work.									
A. CONCEPT										
Engineering Clinic laboratory provides hands-on training for students to develop certain simple real-world products or applications with the help of faculty. It is a team activity consisting of maximum 3 students per team. A list of products or applications will be given. Engineering Clinic - I focus on product development involving Electronics Engineering. Apart from electronic system design the course module has the design and fabrication of Printed Circuit Board (PCB) as well. Each team can choose one or more products for a given application. The students have to design, fabricate and demonstrate the working of the product.										
B. EXECUTION										
Day	Session	Course content / Activity						No. of Periods		
1	S 1	Introduction to Electronics components.						4		
	S 2	Functioning of Electronic components and circuits.						4		
2	S 3	Hands-on Training to design electronic circuits using open-source software.						8		
	S 4	Fabrication of PCB.						4		
3	S 5	Assembling and Soldering of Electronic components in PCB.						4		
	S 6	Testing and Validation of the circuit.						6		
Total							30 Periods			
A list of sample applications/products is attached.										
C. ASSESSMENT										
i. Assessment is done by Internal mode only and there is no End Semester Examination.										
ii. Marks distribution for Internal Assessment is,										
Method	Review I		Review II		Review III		Review IV			
Details	Designing of Electronic circuits using open-source software		Fabrication of PCB		Assembling and Soldering of Electronic components in PCB		Testing, Validation and Demonstration			
Marks	25		25		25		25			
For Product/Application the student team can choose themselves.										
Total: 30 PERIODS										

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Course Outcomes: Upon completion of this course the students will be able to:		BLOOM'S Taxonomy
CO1	Understand the Basics of electronic components	L2- Understand
CO2	Design, Fabrication and Demonstration of the prototype of Electronic product using PCB	L4 - Analyze
CO3	Practice the culture of Innovation and Product Development towards Start-ups in an Institution	L4 - Analyze

Mapping of COs with POs and PSOs														
COs	POs											PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	3	1	2	2		2	2	2		3	3	1
CO2	3	3	3	2	2	1.5		2	2	3		3	3	1
CO3	3	3	3	2	2	1.5		2	3	3		3	3	1
Average	3	3	3	1.6	2	1.6		2	2.3	2.6		3	3	1
1-Low, 2 -Medium, 3-High														

List of sample Applications / Products for Engineering Clinic I

1. Water level indicator in a tank.
2. Automatic solar light circuit.
3. Rain alarm indicator.
4. Fire alarm sensor.
5. LPG gas leakage detector.
6. Air quality measurement.
7. Automatic sanitizer dispenser.
8. Automatic doorbell ringer.
9. Miniature of Home / Buildings / Bridges.
10. Miniature of Hydraulic Jack / Air Pump / Steam power electricity model.

BE23PT806	APTITUDE SKILLS - I	CP	L	T	P	C
		1	0	0	1	0.5
Programme & Branch	COMMON TO ALL B.E. / B.TECH.BRANCHES	Version: 1.1				
Course Objectives:						
1.	To know different methods for faster numerical computations					
2.	To learn Logical reasoning skills.					
	INTRODUCTION(Not for Examination)	1				
Importance	Speed Maths and Logical Reasoning aim to develop quick calculation skills and systematic thinking for efficient problem solving. This course enhances accuracy, speed, and analytical ability required for aptitude tests. Key Areas Where Engineering Students Benefit from Aptitude Skills. 1. Placement and Recruitment Tests 2. Competitive and Higher Studies Exams 3. Professional and Technical Applications					
Real-life Example(s)	1. Rapidly estimating measurements, material requirements, or dimensions during design and construction tasks. 2. Quickly calculating costs, discounts, or taxes while shopping or managing project budgets 3. Using logical reasoning to identify faults, optimize processes, or plan tasks efficiently. 4. Solving aptitude questions on time, such as number series, puzzles, or algebra, during recruitment tests.					
Linkages	Aptitude skills – II, Aptitude skills – III & Aptitude skills – IV					
UNIT-I	SPEED MATHS	6				
	Squaring numbers and multiplying numbers faster than the conventional methods - Finding Square roots of numbers faster - Finding Cube roots faster - Solving simultaneous equations faster than conventional methods.					
UNIT-II	LOGICAL REASONING	8				
	Alphabet and Number Series - Odd Man Out Series - Puzzles - Blood Relations -Seating Arrangement and Ordering - Directional Sense Test .					
	Total	15 Periods				
	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy				
CO1	Apply different techniques for faster calculations.	L3 – Apply				
CO2	Solve mathematical problems by applying logical thinking.	L3 – Apply				

w.e.f, 13/10/2025

Passed in BoS Meeting held on 18/09/2025

Approved in Academic Council Meeting held on 11/10/2025

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Board of Studies

Faculty of Science and Humanities

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REFERENCE BOOKS:		
1.	Aggarwal R. S., "Quantitative Aptitude for Competitive Examinations", S.Chand Publishing Company Ltd(s), 2022.	
2.	Arun Sharma, "How to prepare for Quantitative Aptitude for the CAT" Tata McGraw-Hill Publishing, 2022.	
3.	Praveen R. V., "Quantitative Aptitude and Reasoning" PHI Learning Pvt. Ltd., 2016	

WEB REFERENCES:			
S.No.	Publisher	Website link	Type of Content
1.	Indiabix	https://www.indiabix.com/online-test/aptitude-test/	Tests for Practice
2.	Placement preparation	https://www.placementpreparation.io/quantitative-aptitude/	Tests for Practice
3.	Geeks for geeks	https://www.geeksforgeeks.org/aptitude-for-placements/	Learning Resources and Tests for Practice
4.	Prepinsta	https://prepinsta.com/learn-aptitude/	Learning Resources and Tests for Practice

VIDEO REFERENCES:				
S.No.	Video Details	Name of the Expert	Type of Content	Video Link
1.	YouTube	Aptitude buster	Video Lectures	https://www.youtube.com/watch?v=oA0CxMam_Mg&t=864s
2.	YouTube	Freshersworld.com	Video Lectures	https://www.youtube.com/playlist?list=PLjLhUHPsqNYkcq6YOfiywbTfnvf_TN7i9
3.	YouTube	Triumphant Institute of Management Education Pvt. Ltd.	Video Lectures	https://www.youtube.com/@TIME4BankExams/videos
4.	YouTube	Aptitude buster	Video Lectures	https://www.youtube.com/watch?v=oA0CxMam_Mg&t=864s

Mapping of COs with POs and PSOs														
COs	POs											PSOs		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2													
CO2	2													
Avg.	2.0													
1-Low, 2 -Medium, 3-High.														

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